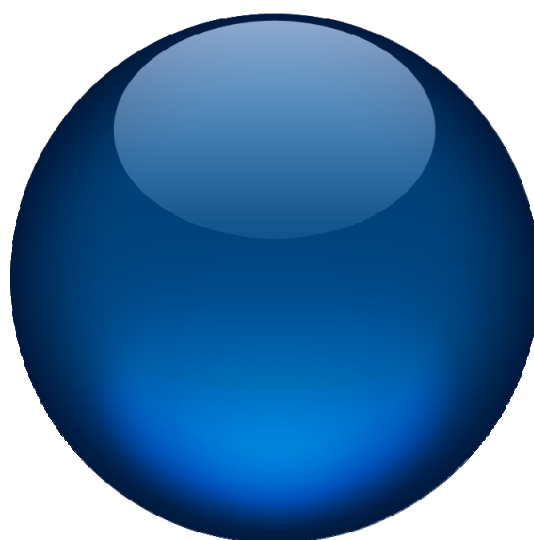




TT801

**REMOTE CONTROL SYSTEM
USER GUIDE**

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Overview of the TT801 User Guide

This document is the User Guide for iXBlue TT801. It must be read and understood prior to using the TT801 system.

The manufacturer shall in no case be held liable for any application or use that does not comply with the stipulations in this guide.

The TT801 User Guide is divided into several parts:

- **Part 1: Introduction:** This part contains a general description of TT801. Details on warranty, iXBlue contacts and technical support are also given.
- **Part 2: Installation:** This part gives the recommendations and the procedures to install the TT801 system.
- **Part 3: Operation:** This part provides the full description of TT801 operation.
- **Part 4: Maintenance**

This part provides the preventive and corrective maintenance for the TT801. It gives some clues as regards troubleshooting.

The abbreviations and acronyms used in this guide are listed hereafter.

A **Table of Contents** is available in the following pages to allow a quick access to dedicated information.

Text Usage

Bold	Bold text is used for items you must select or click in the software. It is also used for the field names used into the dialog box.
<code>Courier</code>	Text in this font denotes text or characters that you should enter from the keyboard, the proper names of disk Drives, paths, directories, programs, functions, filenames and extensions.
<i>Italic</i>	Italic text is the result of an action in the procedures.

Icons



The **Note** icon indicates that the following information is of interest to the operator and should be read.



THE **CAUTION** ICON INDICATES THAT THE FOLLOWING INFORMATION SHOULD BE READ TO FORBID OR PREVENT PRODUCT DAMAGE.



THE **WARNING** ICON INDICATES THAT POSSIBLE PERSONAL INJURY OR DEATH COULD RESULT FROM FAILURE TO FOLLOW THE PROVIDED RECOMMENDATION.

Abbreviations and Acronyms

CAF	Common Acknowledgement Frequency
CIF	Common Interrogation Frequency
CRF	Common Reply Frequency
FR1	Reception frequency number 1 (for RT6xx)
FR2	Reception frequency number 2 (for RT6xx)
FSK	Frequency Shift Keying
FT0	Transmitted Frequency number 0 (the same than CAF)
FT1	Frequency response to FR1 (the same than CIF)
FT2	Frequency response to FR2 (the same than IIF)
FT4	Transmitted Frequency number 4 (the same than PFR)
IIF	Individual Interrogation Frequency
IRF	Individual Reply Frequency
MFSK	Multi-Frequency Shift Keying
MSB	Most Significant Bit
NA	Not Applicable
PFR	Pinger Frequency

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I INTRODUCTION

I.1 Object

This part is the introduction to the TT801 user manual. It contains the description of the TT801 system, of its components, some principles about acoustic remote control, warranty, iXBlue contacts and, technical support.

I.2 Verification of the TT801 Pack Contents

You have just received your equipment in a wooden transit case. Before starting the installation, **we recommend to check the contents of the pack and the equipment immediately on receipt of your TT801 System**. Specifically, you should check that all items are present on delivery and that none has sustained damage.

For that, you will find in the shipping case a packing-list detailing all the shipped items. This packing list had been completed and checked for by iXBlue shortly before shipment.



ON RECEIPT OF THE EQUIPMENT, ITS OVERALL CONDITION SHOULD BE CHECKED AND iXBLUE INFORMED OF ANY DAMAGE SUFFERED DURING THE LOADING OPERATIONS.

IT IS HIGHLY RECOMMENDED TO CONTACT THE INSURANCE COMPANY IN CASE DAMAGE IS FOUND.

I.3 TT801 Overview

The TT801 is a low frequency (8 to 16 kHz) transmitter system of coded acoustic signals to remote control the operating modes of Acoustic Releases and/or Recoverable Transponders. It uses a highly secure and reliable FSK telemetry link in the LF range. TT801 features also a ranging function to help in relocating and monitoring the operation of the subsea Acoustic Releases and Transponders.

The TT801 remote control deck set is a portable unit equipped with a Lead-Acid battery. It can be feed with a continuous or alternative external power source. This external power also allows the charge of internal rechargeable batteries.

Transmission

It is particularly dedicated to transmission of:

- Acoustic release orders towards AR701AE Shallow Water Acoustic Releases
- Remote control of acoustic codes towards RTxx1 and ARxx1 series
- Single Interrogation pulses to OCEANO Transponders

Reception

The receiver section can process:

- Acknowledgement signals of remote control codes
- Information of range and elapsed time
- Verticality and Battery voltage of AR/RT800 and RT600 series by telemetry
- Pressure measurement by telemetry
- Temperature measurement by telemetry

Main features

- The main features are:
 - Portable
 - Range in excess of 10,000m in good environmental conditions
 - Display of data and functions on Liquid Crystal Module
 - Audio indication during the configuration
 - Secure FSK coded commands for absolute multi-path protection
 - Receiver of MFSK (Wide-spread spectrum) and Monochromatic (Tonal) signals
 - Functions: Rangemeter / Pinger receiver / Telemetry / Audio listening
 - Over 30,000 single codes (65,000 couples of codes for AR 701 AE units)
 - Internal rechargeable battery and built-in charger
 - 115-230V AC Mains power supply
 - 24V DC external power supply

I.4 Acoustic Remote Control Principles

Transmission

Depending on the target (acoustic release or release transponders see Table 1 and Table 2), the TT801 can transmit:

- Single tonal pulse for interrogation: Release transponders are generally fitted with two-factory set interrogation frequencies (Individual Interrogate Frequency or Common Interrogate Frequency) ranging from 8.5 to 16 kHz (with 500 Hz steps). The Release Transponder (if 'ENABLE') replies after a turn-around time by sending an acoustic pulse (Common Reply Frequency or Individual Reply Frequency)
- Encoded commands to monitor the operation of Acoustic Release or Release Transponders. iXBlue technique is a FSK-2 states command burst using 8-bit words with 2-state coding. Logical bits '0' and '1' are coded with two different carrier frequencies.

Depending on the target, the transmitted acoustic command may consist of a unique "DIRECT" sequence (single 8-bit word) or an "ARM & EXECUTION" sequence (two 8-bit word burst, the ARM code being equivalent to the address of the commanded subsea unit).

When a command word has been decoded, the AR/RT OCEANO subsea unit transmits an acoustic Acknowledgement pulse. With some commanded functions, the subsea unit transmits a second Acknowledgement pulse when the operation is complete. If the 'DIAGNOSTIC' command was sent, the reply also includes a data using a Pulse-to-Pulse Time Modulation technique. TT801 detects in sequence the acknowledgment pulses and the result / message is posted up on the LCD display.

Table 1 – Correspondence between the 'Interrogate' or 'Reply' frequency names and targets

For:	OIAR701	OIRT6x1	OIAR6x1	OIRT8x1	OIAR8x1	OIRT9xx/RTAx1*	OIAR9x1
Individual Interrogation frequency name	NA	FR2	NA	IIF	NA	IIF	NA
Common Interrogation frequency name	NA	FR1	NA	CIF	NA	CIF	NA
Individual Reply frequency name	NA	FT1	NA	IRF	NA	IRF	NA
Common Reply frequency name	NA	FT2	NA	CRF	NA	CRF	NA
Common Acknowledgement Frequency name	NA	FT0	FT0	CAF	CAF	CAF	CAF
Pinger Frequency	NA	FT4	NA	PRF	PRF	PRF	PRF
Other Acknowledgement frequencies	NA	FT5, FT7 generally set to FT0	NA	NA	NA	NA	NA

* When using an RTA, select RT9 in the TT801. The RTA and RT9 are equally handled by the TT801.

The main commands supported by the different generations of OCEANO Acoustic Releases and Release Transponders are listed hereafter:

- **ARM:** it precedes the commands/orders for OCEANO AR701AE, AR/RT/ET800, AR/RT/ET900 series and operates like a unique address
- **RELEASE:** Used with AR/RT 600 series, OCEANO AR/RT 800 and OCEANO AR701AE to activate the release operation
- **DIAGNOSTIC:** Used with RT600, OCEANO AR/RT800 and AR/RT900 series to ask for information about the verticality and battery voltage of the acoustic release.
- **INT/RANGE:** Used with AR600 series to measure distance from surface to the acoustic release.
- **WINDOW:** Possible use with RT600 series and may precede some RT600 series commands/orders requiring an ARM & EXECUTION sequence. It operates like a unique address and provides the same action as the ARM command. It allows range measurement by sending the WINDOW command

- **ON FR1-FR2:** Used with RT600 series to enable FR1 and FR2 Transponder channels.
- **OFF FR1-FR2-PINGER:** Used with RT600 series to disable FR1 and FR2 Transponder channels and Pinger mode
- **ON FR1-OFF FR2:** Used with RT600 series to enable FR1 and disable FR2 Transponder channels.
- **ON FR2-OFF FR1:** Used with RT600 series to enable FR2 and disable FR1 Transponder channels.
- **PINGER:** Used with RT600 series to activate the Pinger emission
- **PINGER ON:** Used with OCEANO AR/RT/ET800 and AR/RT900 series to activate the Pinger emission and disable the Transponder mode
- **ARM/RANGING:** Allow ranging (distance measurement) on OCEANO AR/RT800 series by sending an ARM single remote control code or a dual ARM & ARM sequence
- **RELEASE with PINGER:** Used with OCEANO AR/RT800 and AR/RT900 series with PINGER: to activate both the release and the pinger emission upon completion of release operation
- **TRANSP. IIF TRANSP.CIF:** Used with OCEANO RT/ET800 and RT900 series to enable IIF and CIF Transponder channels.
- **INHIBITION IIF CIF:** Used with OCEANO RT/ET800 and RT900 series to disable IIF and CIF Transponder channels.
- **TRANSP. IIF INHIB CIF:** Used with OCEANO RT/ET800 and RT900 series to enable IIF and disable CIF Transponder channels
- **TRANSP. CIF INHIB IIF:** Used with OCEANO RT/ET800 and RT900 series to enable CIF and disable IIF Transponder channels
- **PINGER OFF:** Used with OCEANO AR/RT/ET800 and AR/RT900 series to disable and to stop the pinger emission.
- **ACOUSTIC RESET:** Used with OCEANO RT/ET800 and RT900 series. It restores the transponder default configuration of the OCEANO RT/ET800 and RT900 series.
- **TELEMETRY ON-OFF:** Activate the pressure telemetry in transponder mode (IIF).
- **PRESSURE:** Used with RTA series to ask for pressure measurement.
- **TEMPERATURE:** Used with RTA series to ask for temperature measurement.

Table 2 - Supported commands with respect to targets

Target	ARM code	ACK Freq	Supported Commands	ACK Freq.
AR 701 AE	Yes	No	RELEASE	No
RT6x1	Yes or No: Individual choice per command depending on factory set-up	If ARM (FT0) If No ARM, NA	WINDOW ON FR1-FR2 OFF FR1-FR2-Pinger RELEASE 1 DIAGNOSTIC ON FR1-OFF FR2 ON FR2-OFF FR1 PINGER	(FT0) (FT0) (FT0) (FT0 and FT5) (FT0 and FT7) (FT0) (FT0) (FT0 then FT4)
RT8x1 RT9x1 RTAx1	Yes	CAF	ARM/RANGING RELEASE RELEASE WITH PINGER TRANSP.IIF TRANSP.CIF INHIBITION IIF CIF TRANSP.IIF INHIB.CIF TRANSP.CIF INHIB.IIF PINGER ON PINGER OFF DIAGNOSTIC TELEMETRY ON-OFF PRESSURE TEMPERATURE (RTA only) ACOUSTIC RESET	(CAF) (CAF & CAF) (CAF & CAF then PFR) (CAF) (CAF) (CAF) (CAF) (CAF then PFR) (CAF) (CAF and CAF) (CAF) (CAF and CAF) (CAF and CAF) (CAF)
AR6xx	No	NA	INT/RANGE RELEASE	(FT0) (FT0)
AR8xx AR9xx	Yes	(CAF)	ARM/RANGING RELEASE RELEASE WITH PINGER PINGER ON PINGER OFF DIAGNOSTIC	(CAF) (CAF & CAF) (CAF & CAF then PFR) (CAF then PFR) (CAF) (CAF and CAF)

Each of the commands defines above have a code number to enter in the TT801. These code numbers can be found in the provided build sheet of your target (see Appendix C). OCEANO AR/RT/ET800, AR/RT900 and AR701 series are always triggered using a dual word command (ARM and EXECUTION sequence).

AR600 series are always triggered using a single command word (direct command).

RT600 series are usually triggered using a single command word. However some commands can be individually to a dual word command (WINDOW/ARM and EXECUTION sequence). The factory set dual word commands are identified by the “(W)” note on the build sheet.

In the same way, each target has its own interrogation frequencies, type of recurrence, which you have to enter in your TT801. You can find this information in the provided build sheet of your target (see Appendix C).

Reception

Depending on the target,

- The Remote control TT801 receives acknowledgement (CAF for the AR/RT/ET800, AR/RT900 series or FT0 for AR/RT600 series) from the target or not.
- The Remote control TT801 receives some verticality, battery power, distance information, pressure and/or temperature.

The Remote control TT801 is also able to detect the acoustic transmission of free running Pinger and to display the Pinger transmission rate.

Relocation example: Range measurement using Transponder mode (Single pulse)

RT8x1 case

1. TT801 sends a pulse IIF to interrogate a Recoverable transponder (RT).
2. Assuming the transponder channels are active (ENABLE IIF-CIF), the RT will detect the IIF interrogation signal and transmit in turn a reply pulse with the turn around time.
3. TT801 receives a CRF pulse (M-FSK or Tonal code) and displays the range to the RT.

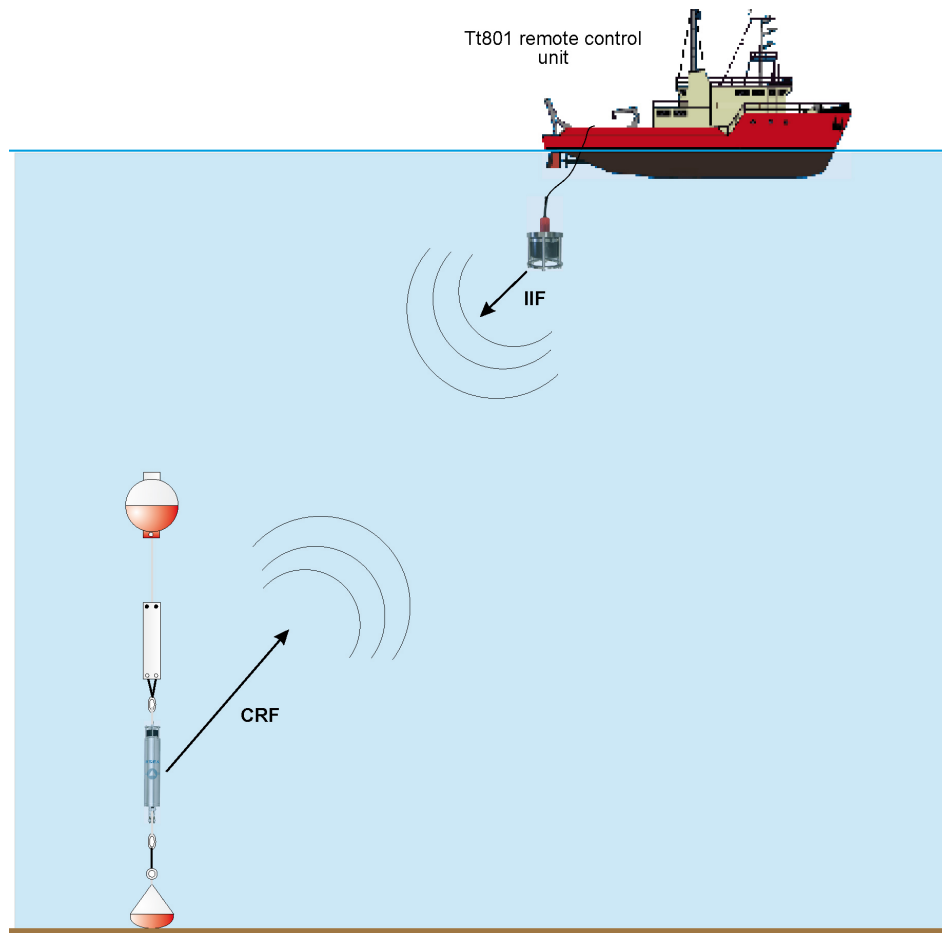


Figure 1 – Relocation example, ranging by a single pulse

Relocation example: 'Ranging' after remote control interrogation

AR8x1 case

1. Set up TT801 to execute ARM & ARM operation with suitable ARM command code and reception parameters.
2. Activate the operation. TT801 transmits once the ARM code.
3. Upon detection of its ARM code, AR8x1 unit transmits an acknowledgement pulse (CAF).
4. TT801 detects CAF pulse and the distance in meters between TT801 and AR8x1.
5. TT801 transmits the second ARM code (Ranging) and the AR8x1 replies in turn by a second CAF pulse, which is also used to measure the distance.

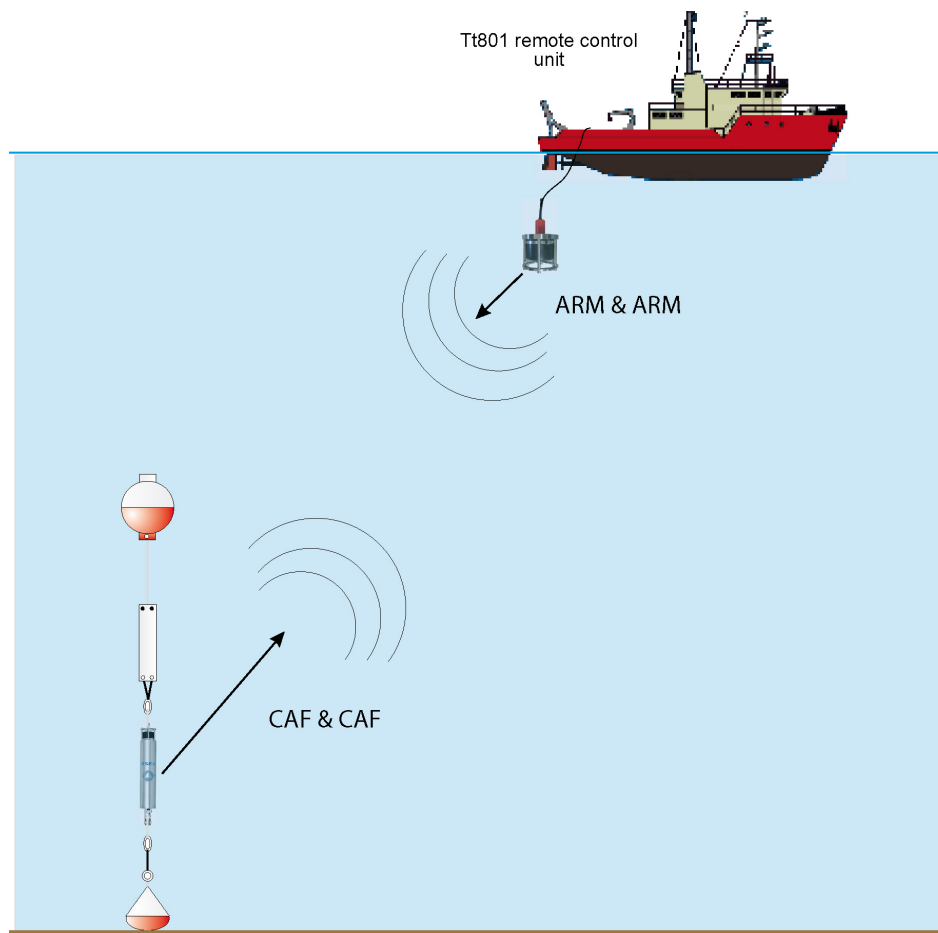


Figure 2 – Relocation example, ranging after interrogation

Mooring Recovery Operation: 'Release + Pinger' command sending example

AR8x1 or RT8x1 case

1. Prior sending any release command, it is strongly recommended to relocate the Acoustic Release (AR800 series for example) by sending RANGING or Diagnostic command.
2. Set up TT801 to execute 'Release + Pinger' operation with suitable ARM + 'Release + Pinger' command codes and reception parameters
3. Activate the command transmission. TT801 transmits once the ARM code.
4. Upon detection of its ARM code, AR8x1 unit transmits an acknowledgement pulse (CAF).
5. TT801 detects CAF pulse and the distance in meters between TT801 and AR8x1.
6. TT801 transmits the 'Release + Pinger' code. Upon receipt of this command, the AR8x1 replies in turn by sending a second CAF pulse, which is also used to measure again the distance.
7. Upon completion of the rotor rotation, the release hook opens and AR8x1 transmits a CAF pulse to confirm execution of release operation.
8. In addition, as pinger mode has been specified, AR8x1 switches to Pinger operation for 3 minutes time by default (configured from the MMI): AR8x1 transmits PFR acoustic pulses at a rate (default is 5 s) at Pinger Frequency. TT801 detects these PFR pulses and displays the Pinger rate.

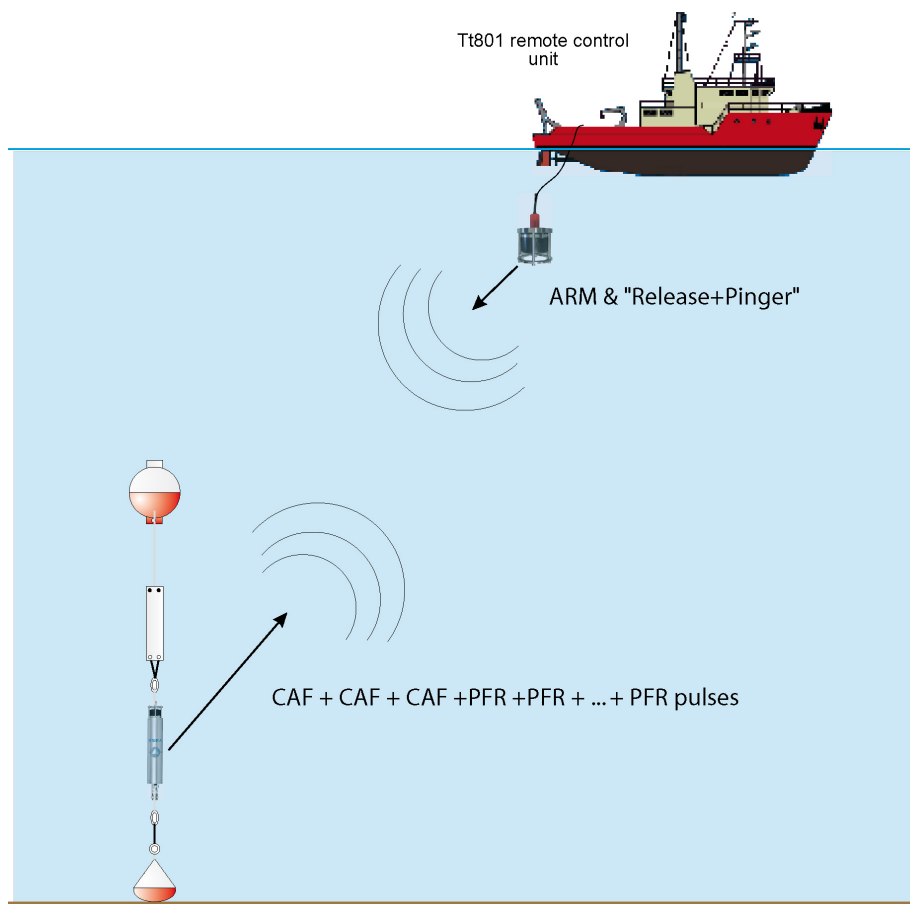


Figure 3 – Relocation example, release and pinger operation

I.5 TT801 Components

I.5.1 PHYSICAL DESCRIPTION

Standard The TT801 unit is supplied as standard with the following accessories (see Figure 4):

- One dunking transducer PET 801 P-30 with 30m long interconnect cable
- One 115-230V AC power cord (2m or 2.5m long depending on standard)
- One 24V DC power cord (2.5m long)
- Audio headphones with associated lead
- One user manual
- One factory approval form (F.A.T.)



Figure 4 - TT801 System Components

Optional Features

The TT801 unit is supplied as standard with the standard dunking transducer PET 801 P-30 with 30m long interconnect cable (as seen in Figure 5).

But, if needed, the following options are available:

- Reinforced transducer PET 661S-TT (pp/n 9400051): Associated to its cable p/n 9300117 (30m long), the PET661S-TT transducer is equivalent to the PET801P-30 standard dunking transducer.



Figure 5 - Reinforced transducer PET 661S-TT

- Acoustic Module AM 661S + 100m long cable on hand reel (P/N 9300102): Use of AM661S Acoustic module is recommended in poor acoustic propagation conditions, in noisy environment or when long cables are required between TT801 cabinet and Dunking Transducer. AM661S use is recommended when the sound velocity profile presents strong gradients near the surface (as it exists in the tropical and equatorial waters for example). Long cable helps also in reducing the impact of vessel noise environment by getting the AM661S as far away as possible from thrusters or propellers.



Figure 6 - Acoustic Module AM661S (left: integral transducer, right: remote transducer)

Audio headphones

This feature helps to give an audio indication of sonar signal reception. Acoustic frequencies received by the dunking transducer are converted into 0-8 kHz audio frequencies. Headphones enable to listen to the acoustic signals in the seawater specifically the CAF Acknowledgements pulses or the PFR pulses when the Pinger mode is activated.

TT801 Front Panel

The TT801 front panel is described in Figure 7.

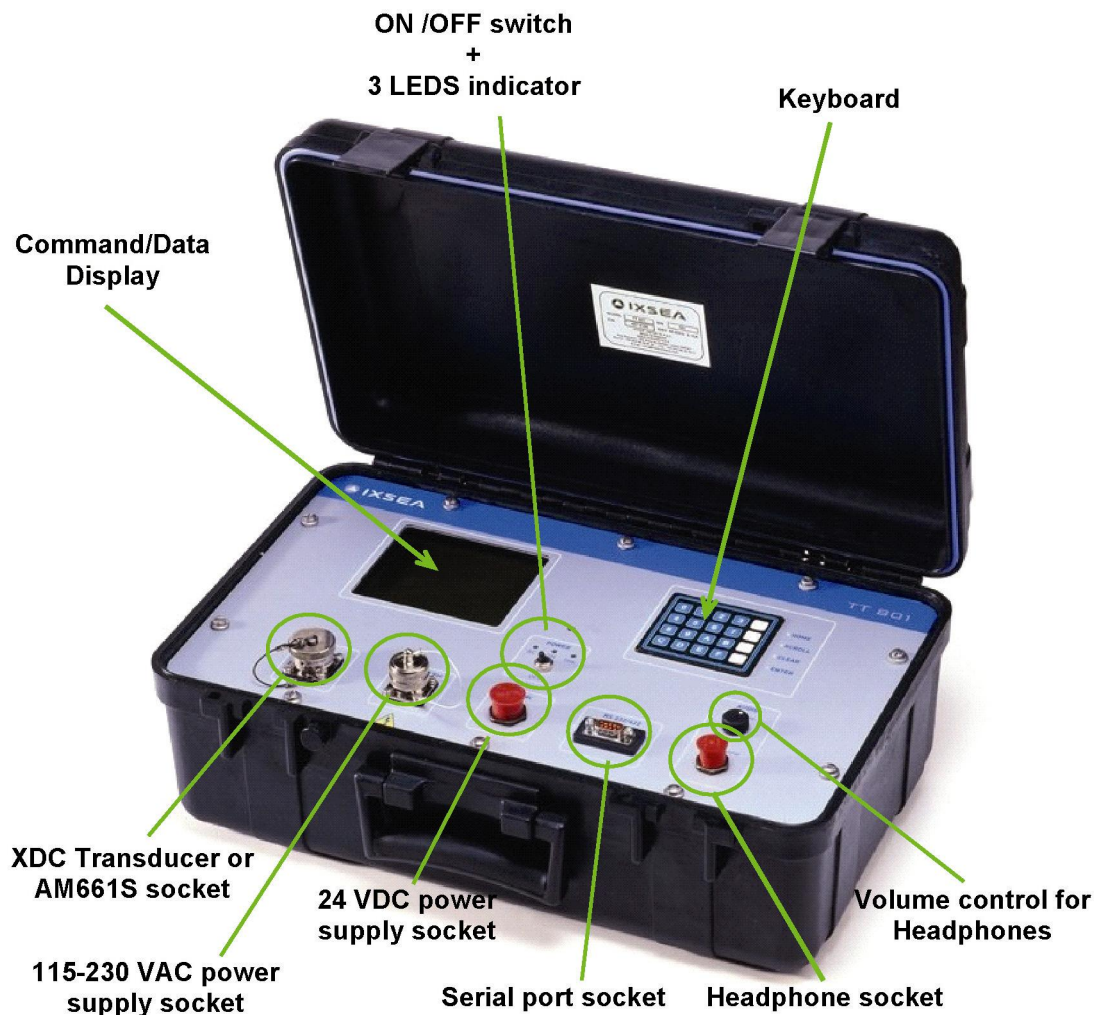


Figure 7 - TT801 Front panel

- **Command/Data Display:** Functions, messages and data are displayed on a 16 lines × 20 characters Liquid Crystal Module. Backlight option makes the reading of data easier in bad visibility conditions.
- **LED indicators:** There are 3 LED indicators on the front panel:
 - ❑ **AC/DC** (yellow) lights on as soon as the TT801 is connected to AC Mains or 24V external power supply
 - ❑ **LOW** (red) lights on when battery is in low charge state
 - ❑ **ON** (green) light on as soon as the power switch is activated to power on the TT801 cabinet
- **Connectors:** Five different types for specific functions:
 - ❑ **XDC** to hook up PET801P dunking transducer or PET661S-TT ruggedized transducer or AM661S acoustic module
 - ❑ **115-230 VAC** universal AC power supply input
 - ❑ **24 VDC** (range 18 to 36 V) DC power supply input
 - ❑ **RS232/422** serial input/output link
 - ❑ **HEADPHONES** (with volume control) for plugging the headset

- **Keyboard:** a 20-key keyboard used for data input and selection of modes. Besides the sixteen usual keys (from 0 to 9 and A to F), four specific keys are available on this keyboard (see section III.3 for details):
 - ❑ **HOME**
 - ❑ **SCROLL**
 - ❑ **CLEAR**
 - ❑ **ENTER**

I.5.2 FUNCTIONAL DESCRIPTION

The TT801 system (Remote Control cabinet) can be broken out as the following:

- One splash proof ABS case
- One front panel
- One Printed Circuit Board
- One power unit fitted with two rechargeable Lead-Acid batteries.

ABS Case Its purpose is to house and protect the front panel and electronics.

Front panel The front panel is used to:

- To plug the various devices and power supply
- To configure, send commands
- To display messages and data

Electronic board The electronic sub-assembly is made of one printed circuit board and uses low power features of CMOS technology and reliable design of iXBlue products. This board contains:

- One DSP for signal processing
- One FPGA to interface with peripherals: EEPROM, Keyboard, Display
- Power supply: general and transmission power; battery charge circuit and surveillance of battery status
- Power amplifier and transfer to the acoustic transducer
- Receiving preamplifier and signal conditioning

Power Unit The power unit is made of:

- Two Lead-Acid battery, YUASA 12V – 3.2Ah, rechargeable (ref.: BT12V3.2AH)
- One 100W AC power supply, universal input 115-230V (ref.: --ALIMEPS100-48S)
- One battery fuse 5A - 250V (ref.: F 5A HPC TDC)
- One AC Mains filter, SCHAFFNER FN2020 1A (ref.: DIFN2020)

One AC Mains fuse 5A - 250V (ref.: F 5A HPC TDC)

I.6 Mechanical/Environmental Characteristics

The mechanical and climatic characteristics are the following:

I.6.1 CABINET

- Splash proof ABS case
- Weight: 9 kg
- Dimensions: 356 × 269 × 157 mm

I.6.2 DUNKING TRANSDUCER

Standard PET 801 P-30 weight + 30 m cable: 4 kg

I.6.3 SHIPPING

Dimensions: 790 × 400 × 390 mm

I.6.4 ENVIRONMENTAL CONDITIONS

- Total weight (with standard PET 801 P-30): 27 kg
- Use inside or outside
- Max. altitude: 6560 ft
- Battery operation temperature range: -5 to +50°C
- Battery charging temperature range: -5 to +30°C
- Max. relative humidity: 80% up to 31°C and linear decrease up to 50% at 50°C
- Pollution level: 3 (conductive pollution)
- Mains power voltage fluctuation within $\pm 10\%$
- Power supply installation category: II

I.7 Electrical Characteristics

I.7.1 FRONT PANEL FEATURES

The front panel includes:

- ON / OFF switch
- SOURIAU 12 way female socket for transducer output (XDC)
- JAEGER 3 way male socket for external 115-230V AC power supply
- SOURIAU 6 way male socket for external 24 VDC power supply
- SOURIAU 4 way female socket for Headphones input
- 9 way “D” male socket for RS232/422 serial link (Provisional)
- ‘AC/DC’ LED indicator for external power supply
- ‘ON’ LED indicator for device’s power-up
- ‘LOW’ LED indicator for low battery voltage
- 20-key keyboard for data and command input
- Liquid Crystal Display 16 lines × 20 characters
- Photo resistor for backlighting command
- Audio buzzer

I.7.2 CONNECTORS

The characteristics of the connectors and plugs are the following:

- **115-230V AC power supply**
 - ❑ Front panel socket: JAEGER 6306 030 06
 - ❑ Cap reference: JAEGER 6307 200 06
 - ❑ Mating plug: JAEGER 6386 030 06
 - ❑ Cable clamp reference: JAEGER 6334 210 06
 - ❑ Wiring:
 - 1 = Chassis Ground
 - 2 = AC Common
 - 3 = AC Input
- **24V DC external power supply**
 - ❑ Front panel socket: SOURIAU 851 07A 10-6 P 50
 - ❑ Mating plug: SOURIAU 851 08 EC 10-6 S 50
 - ❑ Wiring:
 - A-D = Chassis Ground = Earth
 - B-C = (-) Ground Power Supply
 - E-F = (+) Power Supply Input
- **Dunking transducer XDC**
 - ❑ Front panel socket: SOURIAU 851 02 E 14-12 S 50 44
 - ❑ Cap reference: SOURIAU 8500 05 44 B
 - ❑ Mating plug: SOURIAU 851 08 EC 14-12 P 50 44

- ☐ Wiring:
 - A to H: not connected
 - J-L: detection AM / transducer (short-circuit)
 - K: cold point of the transducer
 - M: hot point of the transducer
- **Headphones Input**
 - ☐ Front panel socket: SOURIAU 851 07A 8-4 S 50
 - ☐ Mating plug: SOURIAU 851 08 EC 8-4 P 50
 - ☐ Wiring:
 - A-B= Headphone Signal Output
 - C-D = GND (Electrical Ground)
- **Serial link**
 - ☐ 9 way “D” male socket: FCT FWD F09P
 - ☐ Cap reference: FCT F1042-1PK
 - ☐ Mating plug: 9 way “D” female plug, ITT Cannon ADE9S (or equivalent)
 - ☐ Wiring:

1 =	RS422 TX-	(output from TT801)
2 =	RS232 RX	(output from TT801)
3 =	RS232 TX	(input from TT801)
4 =	RS422 RX+	(input from TT801)
5 =	RS232 Ground	
6 =	RS422 TX+	(output from TT801)
7 =	NC	
8 =	RS422 RX-	(input from TT801)
9 =	NC	

II INSTALLATION

II.1 Object

You find hereafter, all the steps to perform before running the TT801 system.

To...	See...
Know how to handle your TT801 unit	II.2 Handling
Know how to store your TT801 system	II.3 Storage
Get some installation recommendations	II.4 General Recommendations
Know how to get started with TT801 system	II.5 Getting Started with the TT801

II.2 Handling

Do not allow instruments to be dropped or exposed to shock damage, which can cause sensitive internal components to fail.

Control the local environment by preventing personnel stepping into the bight of haphazardly laid cables, which are attached to the instrument.

Do not allow an instrument to block an access or impede normal traffic flows in gangway or on a stairway.

TT801 Deck set remains safe in use when operated and handled with standard operating procedures and written guidelines, therefore in normal operation little threat is posed to the safety of individuals or instruments

Servicing an instrument must always be carried out by competent personnel with appropriate training. Generally, when dismantling equipment, care should always be taken, the disassembly process should be carried out in a clean, free of dust and well-ventilated area.

HIGH VOLTAGES: OPERATION OF THE TT801 DOES NOT PRESENT ANY DANGER TO USERS IN NORMAL USE. IF THE INTERNAL ELECTRONIC PCB IS REMOVED FROM ITS' METAL HOUSING FOR MAINTENANCE, THE USER SHOULD BE AWARE THAT HIGH VOLTAGES ARE PRESENT, PARTICULARLY AT THE TRANSMITTER SECTION WHEN THE DUNKING TRANSDUCER IS CONNECTED TO THE FRONT PANEL.



STORAGE CAPACITORS: TT801 INCORPORATES CAPACITORS, WHICH ARE CAPABLE OF STORING LARGE QUANTITIES OF ELECTRICAL ENERGY. AVOID INADVERTENT DISCHARGE DURING DISMANTLING.

ELECTRONIC COMPONENTS: MOST OF THE CIRCUITS IN THE TT801 USE CMOS LOGIC, WHICH IS EASILY DAMAGED BY STATIC DISCHARGE OR EXCESSIVE SUPPLY VOLTAGE. THE BEST HANDLING FOR ELECTRONICS IS NO HANDLING. FIELD SERVICING CANNOT COMPARE IN EFFICIENCY AND ACCURACY WITH THE ORIGINAL FACTORY ALIGNMENT OR CALIBRATION AND THEREFORE FIELD SERVICING IS SHOULD BE LIMITED TO CHECKING CORRECT OPERATION AND REPAIR BY EXCHANGE OF PCB'S OR MODULES.

II.3 Storage

After the equipment has been inspected for transport damage, the equipment must be placed back in its transport box and stored in a dry area until it is installed.

The equipment must be embarked in its packaging.

BAD STORAGE CONDITIONS MAY VOID CERTAIN CLAUSES OF THE GUARANTEE.

THE LIMITS OF STORAGE TEMPERATURE ARE THE FOLLOWING: $-20^{\circ}\text{C} < T < 50^{\circ}\text{C}$.

IN GENERAL, TO OPTIMIZE PERFORMANCE AND SERVICE LIFE, IT IS RECOMMENDED THAT BATTERIES WHICH ARE TO BE STORED FOR EXTENDED PERIODS OF TIME BE GIVEN A SUPPLEMENTARY CHARGE (FOR EXAMPLE 24 HOURS EVERY THREE MONTHS).



AT HIGH TEMPERATURE THE ELECTRICAL RECOVERABLE CAPACITY OF BATTERY INCREASES AND CONVERSELY AT LOWER TEMPERATURE THE RECOVERABLE CAPACITY OF BATTERY DECREASES. HOWEVER THE SELF-DISCHARGE RATE INCREASES WITH TEMPERATURE. IDEALLY BATTERIES SHOULD BE STORED IN DRY AND COOL CONDITIONS IT IS NECESSARY TO MAINTAIN THE CAPACITY OF BATTERIES BY MORE FREQUENT CHARGES, IF STORED AT A TEMPERATURE LOWER THAN 5°C ,

AVOID EXPOSING THE NEOPRENE BOOT OF THE DUNKING TRANSDUCER TO STRONG SUNLIGHT FOR LONG PERIOD.

II.4 General Recommendations

Prior to any operational use in the field, a complete functional checkout of TT801 Deck set and battery charge should be made.

Whenever it is possible, **it is recommended to operate the TT801 cabinet from internal battery pack with no connection with the AC Mains.**

Surface Cabinet

As TT801 remote control deck set is a portable and splash-proof unit, there is no particular consideration for the location of TT801 cabinet: It can be deployed at the bridge/control room, in shelter, on the rear deck (assuming no water splash) or even in any rubber boat or dingy.

Acoustic transducer

The dunking transducer (PET801, PET661S-TT) or the AM661S transceiver should be deployed to get the neoprene boot of the transducer with clear view, without any obstructions or side reflectors, in the direction of the targets to interrogate.

The dunking transducer should be immersed below the keel of the surface vessel and at minimum depth of 5 to 10 m.

The Dunking transducer (or AM661S) should also be placed as far away as possible from potential acoustic noise and vibrations sources, such as bubbles, thrusters or propellers turbulence, water outlets, auxiliary motors, thrusters, etc...

For a fixed or permanent installation, it is suggested that the Dunking Transducer (or AM661S) should be mounted using an acoustic isolating sleeve (rubber sheeting or similar).

Care should be taken about dissimilar metals when the Dunking Transducer (or AM661S) is not electrically isolated by neutral material. On option, the metal housing may be coated and/or painted and be provided with sacrificial anodes.

For advice on installation methods (pole, sea chest, tow-fish, etc...), please contact the technical department of iXBlue.

Dunking Transducer versus AM661S

Using AM661S Acoustic module is recommended in poor acoustic propagation conditions, in noisy environment or when long cables are required between TT801 cabinet and Dunking Transducer. AM661S use is recommended when the sound velocity profile presents strong gradients near the surface (as it exists in the tropical and equatorial waters for example). Long cable helps also in reducing the impact of vessel noise environment by getting the AM661S as far away as possible from thrusters or propellers.

Headphones

This feature helps to give an audio indication of sonar signal reception and to listen to the acoustic signals in the seawater specifically the CAF Acknowledgements pulses or the PFR pulses when the Pinger mode is activated.

In general human sense of hearing is good detector operating with a lower Signal to Noise ratio than an automatic detector, which may help recognizing the acknowledgment pulses from the Acoustic Release

II.5 Getting Started with the TT801

II.5.1 INSTALLING THE CONNECTIONS

Before getting started, the TT801 should be switched off.

Power supply

TT801 cabinet includes a rechargeable battery, which is normally used to feed the electronic assembly without the need of external power supply.

External power sources are AC-mains or alternately a galvanic isolated DC-voltage.

Whenever possible, it is recommended keeping use of AC-mains to the internal battery charging operation.

- For the AC-Mains, connect the 115-230 AC power supply 2.5 m cord to its JAEGER 3 way socket (see Figure 7)
- For the 24 VDC power supply input, connect the 24VDC power supply cord to the SOURIAU 6 way socket (see Figure 7).

Acoustic transducer

The TT801 unit comes with a standard dunking transducer (PET 801 P-30) fitted with 30m-long cable. Connect the dunking transducer cable to the 'XDC' front panel receptacle.

If using:

- The reinforced transducer PET 661S-TT, connect the 30m long cable p/n 9300117 to the 'XDC' front panel receptacle
- The Acoustic Module AM 661 S, connect the cable p/n 9300102 to the 'XDC' front panel receptacle



For safety reasons, high voltages disappear from the XDC receptacle when NO transducer is hooked up to the 'XDC' front panel receptacle.

Audio Headphones

If you decide to use this feature, connect the headphones cable to the front panel SOURIAU 4 way socket.

II.5.2 INSTALLING THE ACOUSTIC TRANSDUCER

Refer to the section II.4 to choose the best location for the deployment of the Dunking Transducer (or AM661S).

Prior to immersion, clean the hydrophone boot with a soapy solution to remove traces of grease or contaminants. Do not use alcohol or petroleum based solvents on the neoprene boot.

The dunking transducer should be immersed well below the keel of the surface vessel at 5 to 10 m minimum depth or more if possible.

If the Dunking Transducer (or AM661S) is difficult to submerge (strong current, buoyancy, etc...), it is possible to add a dead weight (maximum load should not exceed 100kg):

- Clamped on the electromechanical cable of PET801P-30 thanks to a Chinese finger
- Attached to the protective cage of PET661S-TT or AM661S (transducer end)

An alternative solution is to install the Dunking Transducer on a small tow fish providing a depressor effect, which will help keeping the transducer as deep as possible, navigating in good hydrodynamic conditions and looking downwards to the subsea Acoustic Release deployed on the seafloor.

II.5.3 POWERING THE TT801

Once the various devices are connected and the dunking transducer is deployed in the water, the operator can switch on the TT801 cabinet.

At power up, the configuration of the TT801 is displayed for 3 s.

You can check the serial number of the device, the software version and the battery voltage.

DSP Standby Mode

A DSP standby mode optimizes the autonomy of the unit. This standby mode is activated if no action is applied to the TT801 unit for a 20-min time. Press any key to resume from the standby mode.

After quitting the standby mode, the main screen is posted up. The former display is lost and backlighting returns to default mode (ON).

III OPERATION

III.1 Object

This part contains all the possible operations with the TT801 remote control unit.

To...	See...
Know the characteristics of the TT801 battery	III.2 Battery
Have an overview of the TT801 software	III.3 TT801 Menus Overview
Configure the transmission/reception parameters	III.4 Configuring the Transmission/Reception Parameters
Activate the transmission/reception of acoustic data	III.5 Transmission and Data Reception Display
Activate the release	III.6 Activating the Release
Activate the Pinger frequency detection	III.7 Activating the Pinger Frequency Detection
Choose the interface language, turn off/on the display backlight, get the battery voltage, have a beep command	III.8 Activating the Options
Using the serial link	III.9 Using the Serial Link

III.2 Battery

When running on internal battery, the TT801 remote control unit is **operational for about 60 hours** (without backlighting mode). A red LED indicator (**LOW**) lights on as soon as battery reaches the low charge state.

III.3 TT801 Menus Overview

Keyboard Details

Function and data are displayed on the LCD display area (see Figure 7). The 20-key keyboard is used for data input and selection of modes.

Besides the sixteen usual keys (from 0 to 9 and A to F), four specific keys are available on this keyboard. Their uses are described in Table 3.

Table 3 - Specific keyboard keys

Key	Use
HOME	Whatever the situation in the TT801 menu you are, pressing this key leads to the main screen. This key is also used to leave a function of the menu and cancel the outstanding input of data.
SCROLL	When several screens or data are available in a same menu, press this key to scroll screens or data, in particular when programming: • Type of the instrument • Type of transmission • Transmitted frequency • Type of recurrence for transmissions IIF, CIF, FR1, FR2 • Type of transmitted code • Type of received code • Type of receiver sequence • Type of expected received data • Configuration (associated to an instrument serial number)
CLEAR	This key is used to clear the writing in progress field.
ENTER	Press this key in order to confirm the selection of a command or data input.



Wrong command or key press is accompanied by a “BEEP” if Audio mode is active.

Available Menus

The available menus are sum-up in the following figure of the software architecture:

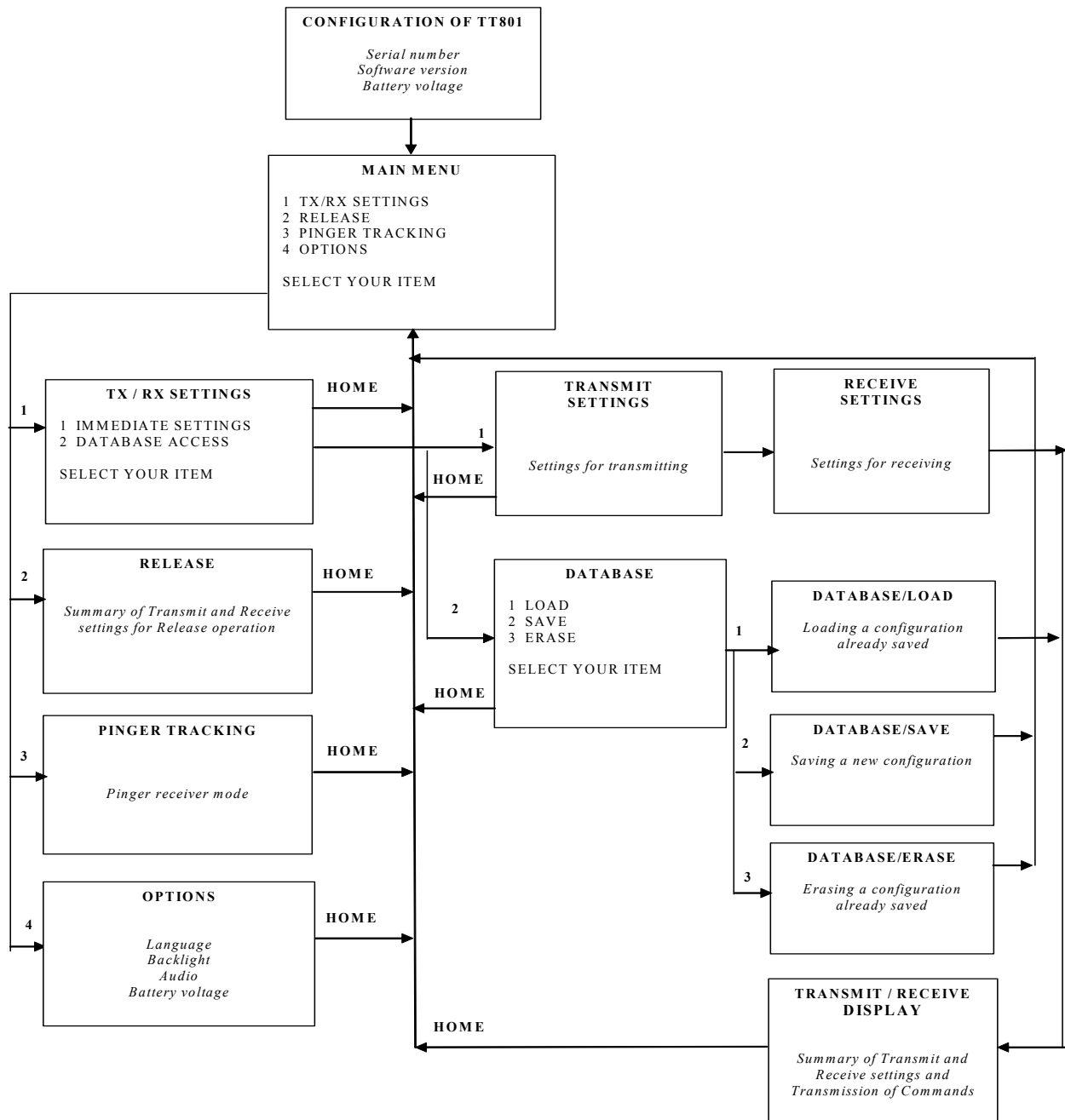


Figure 8 - Menus Architecture

Main Menu

You access this menu after switching on the TT801 unit and once the serial number, software version and battery voltage have been displayed for a few seconds.

- Pressing 1 key allows to set the transmission and the reception parameters: from this menu, you can program transmission and reception parameters of TT801 without saving the configuration (“IMMEDIATE TRANSMIT” submenu) or to get to the “DATABASE ACCESS” sub-menu in order to:
 - ❑ Load a pre-programmed configuration with the ID number of an instrument
 - ❑ Save a new configuration, that is recording codes and parameters linked to the operation of TT801 transmitter and receiver
 - ❑ Delete an existing configuration
- Pressing 2 key allows displaying programmed parameters and transmitting codes for release.
- Pressing 3 key allows displaying Pinger tracking menu.
- Press 4 key allows modifying the parameters of the remote control unit (backlight, language...).

III.4 Configuring the Transmission/Reception Parameters

III.4.1 DEFINITIONS

Immediate Settings

This menu is used to set up transmission and reception parameters without saving the configuration. The access to this sub-menu is allowed by pressing **1** key in the menu **TX / RX settings**.

Database Access

This menu is used for the management of a Database of instruments to be triggered with TT801 cabinet. It helps to Save / Load / Erase the transmission and reception parameters. The access to this sub-menu is allowed by pressing **2** key in the menu **TX / RX settings**.

Transmission Parameters

Monochromatic Interrogation

The user can configure a monochromatic (tonal) single Interrogation pulse; in this case, the parameters to set are the following:

- The TYPE: RT6xx, RT8xx or RT9xx
- The TRANSMIT TYPE:
 - Either the Individual Interrogation Frequency, IIF or the Common Individual Frequency CIF for the RT8xx and RT9xx
 - Either the FR1 or FR2 for the RT6xx
- The transmit frequency FREQ: 8.5 to 16 kHz with incremental step of 0.5 kHz
- The recurrence type: Manual (MANUAL) or automatic (AUTO)
- The transmission rate (RATE), between 2 and 31s, if AUTO has been chosen for the recurrence

Commands with FSK transmission

The user can configure a FSK command for the transmission; in this case, the following parameters have to be set:

- The TYPE: RT6xx, AR6xx, AR7xx, AR8xx or RT8xx, AR9xx or RT9xx
- The TRANSMIT TYPE: ARM + CMD (for RT6xx, AR7xx, AR8xx or RT8xx, AR9xx or RT9xx) or CMD ONLY (RT6xx and only for AR6xx)



Each command code for RT6xx unit can be individually configured with **CMD only** sequence or **ARM & CMD** sequence. If the **ARM & CMD** sequence is required, the command code, mentioned on the RT6xx built sheet, is followed by the **(W)** label.

- The CODE ARM (only for ARM + CMD sequence) and then CMD: these are the hexadecimal ARM and COMMAND codes of the commanded function to initiate
- The type of command code to be sent to the beacon:
 - For RT8xx, AR8xx, RT9xx, AR9xx: Release, Release + Pinger, Pinger ON, Telemetry (Diagnostic, Pressure, Temperature), Other CMD (other function)
 - For RT6xx: Release, Pinger ON, Telemetry (Diagnostic), Other CMD (other function)
 - For AR7xx: Release
 - For AR6xx: Release, other CMD (for RANGING/INT)

Reception Parameters

The type of reception parameters to configure depends on the type of transmission parameters. See Table 4, Table 5, or Table 6 for the correspondence.

Table 4 - For a Monochromatic Transmission Type (single pulse)

Beacon type	Transmission Type	Reception parameters to set
RT/ET8xx RT9xx/RTAxx	IIF	Turn Around Time: this is the delay between the beacon detection of interrogation and acoustic reply CRF type: Mono or MSFK. CRF Frequency or CODE: Tonal Frequency from 8.0 to 16 kHz, by 500 Hz step (if Mono) or MFSK code# from 0 to 23. RANGING: it is the type of information to display, Distance or Elapsed Time.
RT/ET8xx RT9xx RTAx1	CIF	Turn Around Time: this is the delay between the beacon detection of interrogation and acoustic reply IRF type: Mono or MSFK. IRF CODE or Frequency (8 to 16 kHz). RANGING: it is the type of information to display, Distance or Elapsed Time.
RT/ET6xx	FR1	Turn Around Time: this is the delay between the beacon detection of interrogation and acoustic reply. FT1 type: Mono (only). FT1 Frequency: 8 to 16 kHz, by 500 Hz step. RANGING: it is the type of information to display, Distance or Elapsed Time.
RT/ET6xx	FR2	Turn Around Time: this is the delay between the beacon detection of interrogation and acoustic reply. FT2 type: Mono (only). FT2 Frequency: 8 to 16 kHz, by 500 Hz step. RANGING: it is the type of information to display, Distance or Elapsed Time.

Table 5 - For ARM+CMD Commands Transmission

Beacon type	Command	Reception parameters to set
AR7xx	Release	None
8xx, 9xx, Axx series	Release Telemetry (Diagnostic, Pressure, Temperature) Other CMD	CAF type: MONO (or MFSK for Diagnostic) and Frequency (or CODE). RANGING: it is the type of information to display, Distance or Elapsed Time.
8xx, 9xx, Axx series	Release + Pinger, Pinger ON	FT0 type: MONO or MFSK and CODE Frequency (8 to 16 kHz). PFR (Pinger FRequency) type: MONO (or MFSK) and Frequency (8 to 16 kHz) or MFSK Code. RANGING: it is the type of information to display, Distance or Elapsed Time.
RT6xx series	Release Telemetry (Diagnostic) Other CMD	FT0 type: MONO (Only) and Frequency (8 to 16 kHz) RANGING: it is the type of information to display, Distance or Elapsed Time.
RT6xx series	Pinger ON	FT0 type: MONO and Frequency (8 to 16 kHz). FT4 (Pinger Frequency) type: MONO and Frequency (8 to 16 kHz). RANGING: it is the type of information to display, Distance or Elapsed Time.

Table 6 - For CMD ONLY commands Transmission

-+	Command	Reception parameters to set
AR6xx series	Release Other CMD (INT / Ranging)	FT0 type: MONO and Frequency (8 to 16 kHz). RANGING: it is the type of information to display, Distance or Elapsed Time.
RT6xx series	Release Telemetry (Diagnostic) Other CMD	FT0 type: MONO and Frequency (8 to 16 kHz). RANGING: it is the type of information to display, Distance or Elapsed Time.
RT6xx series	Pinger ON	FT0 type: MONO and Frequency (8 to 16 kHz). FT4 (Pinger Frequency) type: MONO and Frequency (8 to 16 kHz). RANGING: it is the type of information to display, Distance or Elapsed Time.

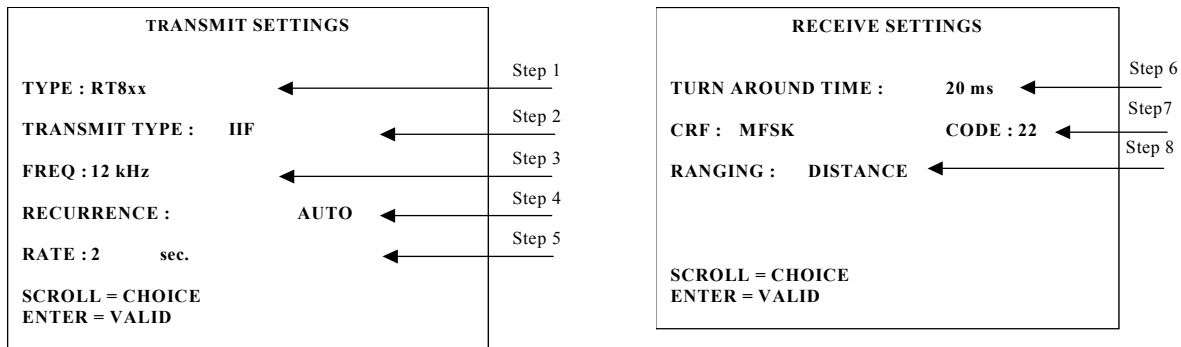
III.4.2 CONFIGURING THE Tx/Rx PARAMETERS WITHOUT SAVING THE CONFIGURATION

The access to this sub-menu is allowed by pressing **1** key in **TX / RX settings** menu.

Depending on the type of instrument and which function to monitor, the operator has to run the following procedures.

III.4.2.1 Monochromatic Transmission/Interrogation Case (RT/ET6xx, RT/ET8xx, RT9xx)

The following figure sums up the procedure to perform:



The access to the **IMMEDIATE TRANSMIT** sub-menu is allowed by pressing **1** key in the menu **TX / RX settings**. *The TRANSMIT SETTINGS display opens.*

To configure the various parameters according to the built sheet of the instrument to command, perform the following procedure:

Procedure

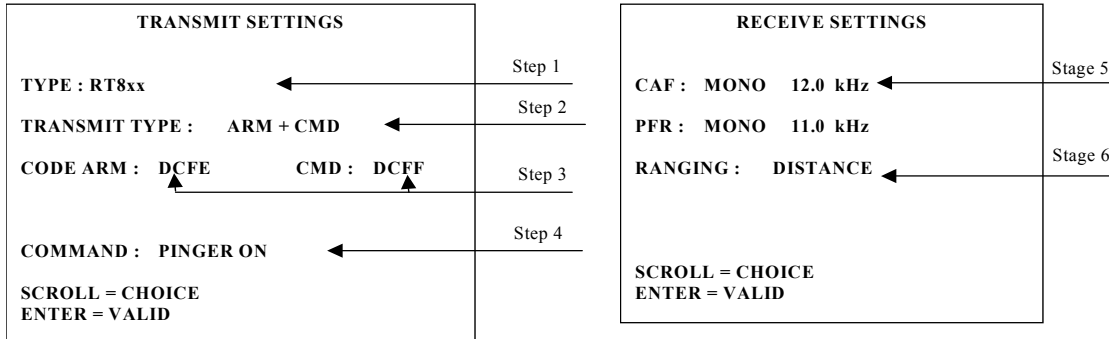
Step	Action
1.	Select the type (TYPE) of the instrument to be actuated using SCROLL key (RT6xx or RT8xx or RT9xx) and confirm the selection by pressing ENTER key.
2.	Select the type of transmission (TRANSMIT TYPE) using SCROLL key: • IIF, CIF (for RT8xx and RT9xx), • FR1, FR2 (for RT6xx), and then confirm the selection by pressing ENTER key.
3.	Select the transmit frequency (FREQ) (from 8.5 up to 16kHz by 0.5kHz increment) using SCROLL key then confirm the selection by pressing ENTER key.
4.	Select the type of recurrence MANUAL or automatic (AUTO) using SCROLL key and confirm the selection by pressing ENTER key. <i>If Manual is chosen, the RECEIVE SETTINGS display opens (skip to step 6).</i>
5.	If AUTO has been chosen in the previous step, then you have to enter the transmission rate (RATE) between 2 and 31s using the numerical keys then press ENTER key. <i>Then, the RECEIVE SETTINGS display opens.</i>
6.	Enter the TURN AROUND TIME (Refer to beacon built sheet) i.e., the turn-around delay of the beacon (between 0 and 200 ms), and confirm by pressing ENTER key. Use CLEAR key to erase a wrong capture.

Step	Action
7.	Enter the TT801 receive parameters according to the reply parameters of subsea transponder: - If the type of transmission/interrogation is IIF (or FR2 for RT6xx), you must configure the CRF (FT2 for RT6xx): - Choose the receive type (MONO or MFSK) using the SCROLL key then press ENTER key to confirm. Note: If IIF is set to MONO the RECEIVING SETTINGS display posts up the PRESSURE MEAS. question after RANGING live. - Choose the frequency (8 to 16 kHz) by using the SCROLL key or the numerical key for the MFSK code number (0 to 23) of the CRF. Press ENTER key to confirm. - If the type of transmission/interrogation is CIF (or FR1 for RT6xx), you must configure the IRF (or FT1 for RT6xx): - Choose the receive type (MONO or MFSK) using the SCROLL key then press ENTER key to confirm. Choose the frequency (8 to 16 kHz) by using the SCROLL key or the numerical key for the MFSK code number (0 to 23) of the CRF. Press ENTER key to confirm.
8.	Select the type of the information to display (RANGING): DISTANCE or Elapsed TIME using SCROLL key and press ENTER key to confirm. <i>Then if the PRESSURE MEAS. is not displayed the TT801 opens the TX / RX display for sending the command in water.</i> To proceed further, skip to section III.5.
9.	If the message PRESSURE MEAS. is posted up, select: ON if a pressure telemetry is expected from the subsea instrument being interrogated (i.e. two successive replies from the subsea instrument) OFF if no pressure telemetry expected from the subsea instrument Then, the TT801 opens the Tx/Rx display for sending the command in water. To proceed further, skip to section III.5.
10.	End of procedure.

III.4.2.2 FSK Transmission Cases

ARM + CMD commands (RT9xx, AR9xx, RT8xx, AR8xx, RT6xx, AR7xx)

The following figure sums up the procedure to perform:



The access to the 'Immediate' settings sub-menus is allowed by pressing **1** key in the menu **TX / RX settings**. The **TRANSMIT SETTINGS** display opens.

To configure the various parameters according to the built sheet of the instrument to command, perform the following procedure:

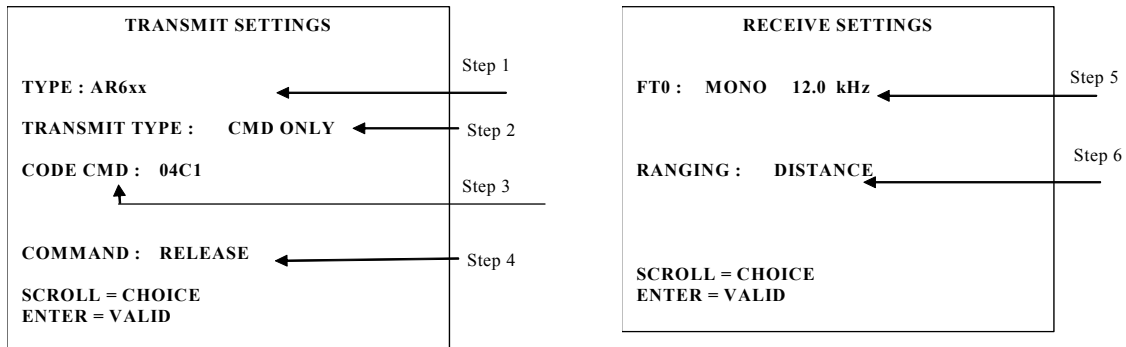
Procedure

Step	Action
1.	Select the type (TYPE) of the instrument to be actuated using SCROLL key and confirm the selection by pressing ENTER key.
2.	Select the type of transmission (TRANSMIT TYPE) using SCROLL key, according to individual built sheet of the beacon and confirm the selection by pressing ENTER key. The type of transmission is normally ARM + CMD .
3.	Enter the hexadecimal ARM code using the alpha-numerical keys and confirm by pressing ENTER key, then enter the hexadecimal command code (2 nd byte only) and confirm by pressing ENTER key.
4.	Indicate to the TT801 which type of command will be sent to the instrument: - For AR/RT8xx, AR/RT9xx: Release, Release + Pinger, Pinger ON, Diagnostic, Pressure, Temperature, Other CMD - For RT6xx: Release, Pinger ON, Telemetry (Diagnostic), Other CMD - For AR7xx: Release Selection is made with SCROLL key and confirmation with ENTER key. The RECEIVE SETTINGS display opens according to the command type, except for AR7xx unit that does not transmit any acoustic reply. With AR701, TT801 operation skips to step 7.

Step	Action
5.	Configure the CAF (or FT0 for RT6xx series) receiver: - Choose the receive type (MONO or MFSK) using the SCROLL key then press ENTER key to confirm. - Choose the frequency (8 to 16 kHz) by using the SCROLL key or the numerical key for the MFSK code number (0 to 23) of the CAF. Press ENTER key to confirm. if you have chosen 'Release + Pinger' (AR/RT8xx and AR/RT9xx) or 'Pinger ON' function, you also have to define the Pinger FRequency PFR (FT4 for RT6xx): - Choose the receive type (MONO or MFSK) using the SCROLL key then press ENTER key to confirm. - Choose the frequency (8 to 16 kHz) by using the SCROLL key or the numerical key for the MFSK code number (0 to 23) of the PRF. Press ENTER key to confirm.
6.	Select the type of the information to display (RANGING): DISTANCE or Elapsed TIME using SCROLL key and press ENTER key to confirm. <i>Then if the PRESSURE MEAS. is not displayed the TT801 opens the TX / RX display for sending the command in water.</i> To proceed further, skip to section III.5.
7.	End of procedure.

CMD ONLY commands (All AR6xx or RT6xx for function without dual code sequence)

The following figure sums up the procedure to perform:



Procedure The access to the 'Immediate' settings sub-menus is allowed by pressing 1 key in the menu TX / RX settings. The TRANSMIT SETTINGS display opens.
To configure the various parameters, perform the following procedure:

Procedure

Step	Action
------	--------

- | | |
|----|---|
| 1. | Select the type (TYPE) of the instrument to be actuated (All AR6xx or some RT6xx) using SCROLL key and confirm the selection by pressing ENTER key.
<i>The TRANSMIT Type is CMD ONLY.</i> |
|----|---|

- | | |
|----|---|
| 2. | Select the type of transmission (TRANSMIT TYPE), according to individual built sheet: <ul style="list-style-type: none"> With AR6xx no other choice than CMD ONLY. With RT6xx, if function is to be activated by a single direct word sequence, select CMD ONLY by using SCROLL key. Then confirm the selection by pressing ENTER key. |
|----|---|

- | | |
|----|--|
| 3. | Enter the hexadecimal code (CODE CMD) using the alphanumerical keys and confirm by pressing ENTER key. |
|----|--|

- | | |
|----|---|
| 4. | Indicate to the TT801 which type of command code will be sent to the beacon <ul style="list-style-type: none"> For RT6xx: Release, Pinger ON, Telemetry (Diagnostic), Other CMD For AR6xx: Release or Other CMD (for INT/Ranging). Selection is made with SCROLL key and confirmation with ENTER key.
The RECEIVE SETTINGS display opens, according to the command type. |
|----|---|

- | | |
|----|---|
| 5. | Configure the FT0 receiver of TT801: <ul style="list-style-type: none"> The receive type is MONO only (No other choice). Press ENTER key to confirm. Choose the FT0 frequency (8 to 16 kHz) by using the SCROLL key. Press ENTER key to confirm. if you have chosen ' Pinger ON ' command (RT6xx), you also have to define the receiver channel for the Pinger Frequency FT4 : <ul style="list-style-type: none"> The receive type is MONO only (No other choice). Press ENTER key to confirm. Choose the FT4 frequency (8 to 16 kHz) by using the SCROLL key. Press ENTER key. |
|----|---|

Step	Action
6.	Select the type of the information to display (RANGING): DISTANCE or Elapsed TIME using SCROLL key and press ENTER key to confirm. <i>Then TT801 opens the TX / RX display for sending in water the command.</i> To proceed further, skip to section III.5.
7.	End of procedure.

III.4.3 LOADING A PRE-PROGRAMMED CONFIGURATION

The access to this sub-menu is allowed by pressing **2** key in **TX / RX settings** menu and then by pressing **1** key in the **DATABASE** menu. Then, the DATABASE/LOAD display opens with the first instrument available in the database.

DATABASE / LOAD		Step 1
		
IDENT :	001	TYPE : RT8xx
TRANSMIT TYPE :		ARM + CMD
CODE ARM :	EF FE	CMD : EF CD
COMMAND :	OTHER CMD	
SCROLL = CHOICE		
ENTER = VALID		

Procedure

Step	Action
------	--------

1. Select the new device to be loaded with **SCROLL** key.

TT801 displays the main parameters of the instruments selected.

To make correction of the ID, use the **CLEAR** key:

A blank window opens with requirement for an ID number.

You must then type in an existing saved ID number and confirm by pressing the **ENTER** key to display the configuration of the selected instrument. If the selection is not valid, the message **"ID NOT REGISTERED"** is prompted and you are required to exit ("HOME" key) or to type in a new ID number.

*Then, press **ENTER** key to confirm the selection; the Transmit/Receive display opens automatically. To proceed further, skip to section III.5.*

2. End of procedure.



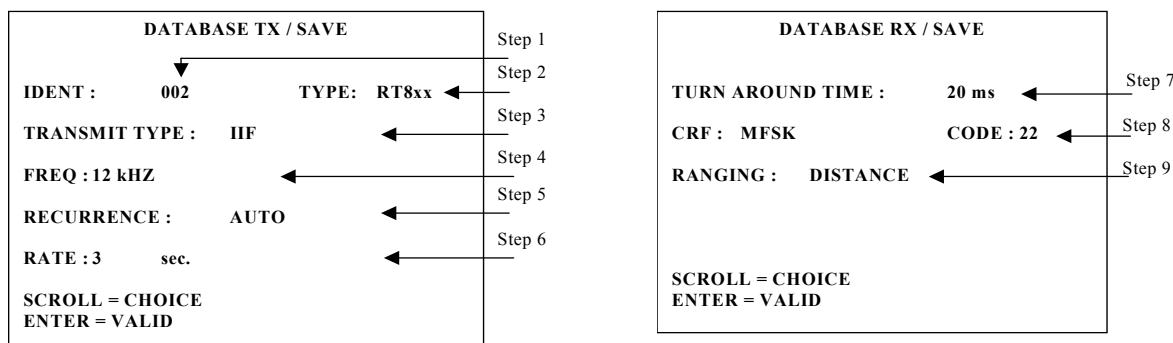
If several devices have been registered with the same ID number, the first one saved with this ID number will be loaded.

III.4.4 SAVING A NEW CONFIGURATION

The access to this sub-menu is allowed by pressing **2** key in **TX / RX settings** menu and then by pressing **2** key in the **DATABASE** menu. The DATABASE TX/SAVE display opens first. Proceed with parameter entry as described hereafter, according to the action to execute (Single pulse Transmission or FSK commands).

III.4.4.1 Monochromatic Transmission/Interrogation Case (RT6xx, RT8xx, RT9xx)

The following figure sums up the procedure to perform:



Procedure

To configure and save the various parameters according to the built sheet of the instrument to command, perform the following procedure:

Procedure

Step	Action
------	--------

1. Select (key-in) the Device ID number (**IDENT**). To make correction you can use the **CLEAR** key. Then press **ENTER** key to confirm.
2. Select the type (**TYPE**) of the instrument to be actuated using **SCROLL** key (**RT6xx** or **RT8xx** or **RT9xx**) and confirm the selection by pressing **ENTER** key.
3. Select the type of transmission (**TRANSMIT TYPE**) using **SCROLL** key:
 - **IIF**, **CIF** (for RT8xx and RT9xx)
 - **FR1**, **FR2** (for RT6xx)
and then confirm the selection by pressing **ENTER** key.
4. Select the transmit frequency (**FREQ**) (from 8.5 up to 16kHz by 0.5kHz increment) using **SCROLL** key then confirm the selection by pressing **ENTER** key.
5. Select the type of recurrence **MANUAL** or automatic (**AUTO**) using **SCROLL** key and confirm the selection by pressing **ENTER** key.

If **Manual** is chosen, the **DATABASE RX/SAVE** display opens (skip to step 7).
6. If **AUTO** has been chosen in the previous step, then you have to enter the transmission rate (**RATE**) between 2 and 31s using the numerical keys then press **ENTER** key.

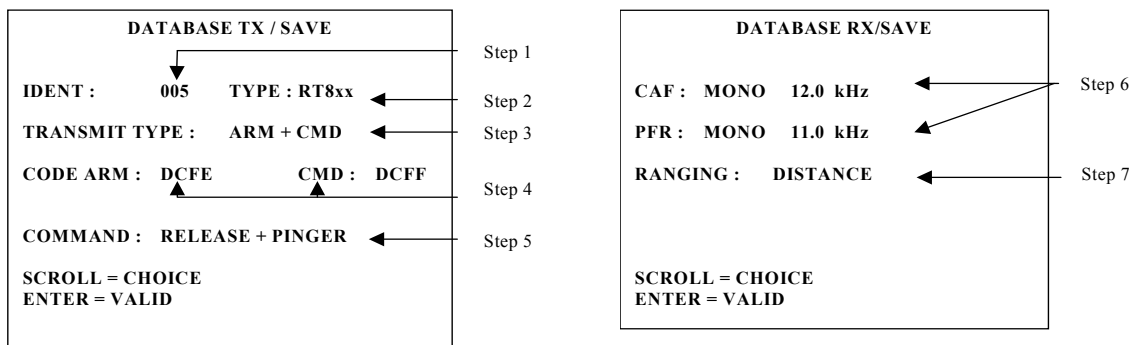
The **DATABASE RX/SAVE** display opens.

Step	Action
7.	Enter the TURN AROUND TIME (Refer to beacon built sheet) i.e., the turn-around delay of the beacon (between 0 and 200 ms), and confirm by pressing ENTER key. Use CLEAR key to erase a wrong capture.
8.	- Enter the TT801 receive parameters according to the reply parameters of subsea transponder: - If the type of transmission/interrogation is IIF (or FR2 for RT6xx), you must configure the CRF (FT2 for RT6xx): - Choose the receive type (MONO or MFSK) using the SCROLL key then press ENTER key to confirm. Note: If IIF is set to MONO the RECEIVING SETTINGS display posts up the PRESSURE MEAS. question after RANGING live. - Choose the frequency (8 to 16 kHz) by using the SCROLL key or the numerical key for the MFSK code number (0 to 23) of the CRF. Press ENTER key to confirm. - If the type of transmission/interrogation is CIF (or FR1 for RT6xx), you must configure the IRF (or FT1 for RT6xx): - Choose the receive type (MONO or MFSK) using the SCROLL key then press ENTER key to confirm. Choose the frequency (8 to 16 kHz) by using the SCROLL key or the numerical key for the MFSK code number (0 to 23) of the CRF. Press ENTER key to confirm.
9.	Select the type of the information to display (RANGING): DISTANCE or Elapsed TIME using SCROLL key and press ENTER key to confirm. Upon completion of the operation, TT801 displays the message “STORAGE COMPLETED” and return to the <i>MAIN MENU</i> display.
10.	If the message PRESSURE MEAS. is posted up, select: ON if a pressure telemetry is expected from the subsea instrument being interrogated (i.e. two successive replies from the subsea instrument) OFF if no pressure telemetry expected from the subsea instrument Then, upon completion of this operation, if PRESSURE MEAS. is not displayed the TT801 displays the message STORAGE COMPLETED and returns to the MAIN MENU display.
11.	End of procedure.

III.4.4.2 FSK Transmission Cases

ARM +CMD commands (RT9xx, AR9xx, RT8xx, AR8xx, some RT6xx, AR7xx)

The following figure sums up the procedure to perform:



To configure the various parameters according to the build sheet of the instrument to command, perform the following procedure:

Procedure

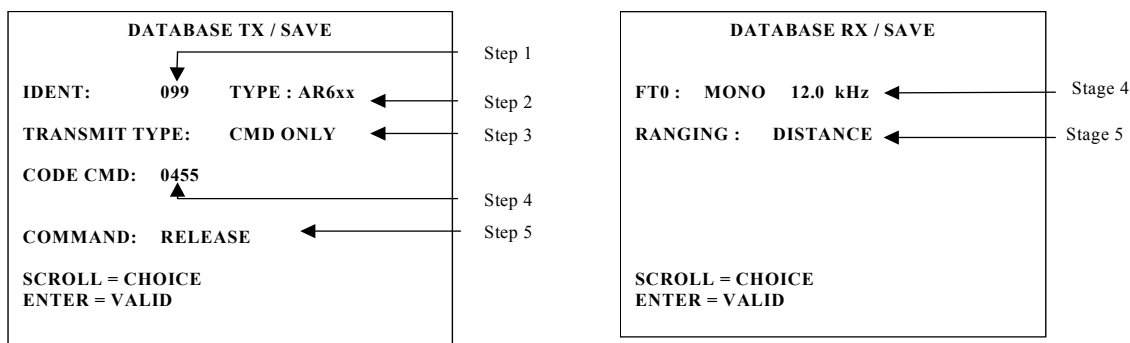
Step	Action
1.	Select (key-in) the Device ID number (IDENT). For entry correction you can use the CLEAR key. Then press ENTER key to confirm.
2.	Select the type (TYPE) of the instrument to be actuated using SCROLL key and confirm the selection by pressing ENTER key.
3.	Select the type of transmission (TRANSMIT TYPE) using SCROLL key, according to individual built sheet of the beacon and confirm the selection by pressing ENTER key. The type of transmission is normally ARM + CMD .
4.	Enter the hexadecimal ARM code using the alphanumerical keys and confirm by pressing ENTER key, then enter the hexadecimal command code (2 nd byte only) and confirm by pressing ENTER key.
5.	Indicate to the TT801 which type of command will be sent to the instrument: - For AR/RT8xx and AR/RT9xx: Release, Release + Pinger, Pinger ON, Diagnostic, Pressure, Temperature, Other CMD - For RT6xx: Release, Pinger ON, Telemetry (Diagnostic), Other CMD - For AR7xx: Release Selection is made with SCROLL key and confirmation with ENTER key.

The **DATABASE RX/SAVE** display opens, except for AR7xx unit which does not transmit any acoustic reply. For AR7X, TT801 skips to step 8.

Step	Action
6.	Configure the CAF (or FT0 for RT6xx series) receiver: - Choose the receive type (MONO or MFSK) using the SCROLL key then press ENTER key to confirm - Choose the frequency (8 to 16 kHz) by using the SCROLL key or the numerical key for the MFSK code number (0 to 23) of the CAF. Press ENTER key to confirm if you have chosen 'Release + Pinger' (AR/RT8xx only) or 'Pinger ON' function, you also have to define the Pinger FRequency PFR (FT4 for RT6xx): - Choose the receive type (MONO or MFSK) using the SCROLL key then press ENTER key to confirm Choose the frequency (8 to 16 kHz) by using the SCROLL key or the numerical key for the MFSK code number (0 to 23) of the PRF. Press ENTER key to confirm
7.	Select the type of the information to display (RANGING): DISTANCE or Elapsed TIME using SCROLL key and press ENTER key to confirm. Upon completion of the operation, TT801 displays the message "STORAGE COMPLETED" and return to the MAIN MENU display.
8.	End of procedure.

CMD ONLY commands (All AR6xx or RT6xx for function without dual code sequence)

The following figure sums up the procedure to perform:



To configure the various parameters according to the built sheet of the instrument to command, perform the following procedure:

Procedure

Step	Action
1.	Select (key-in) the Device ID number (IDENT). For entry correction you can use the CLEAR key. Then press ENTER key to confirm.
2.	Select the type (TYPE) of the instrument to be actuated (All AR6xx or some RT6xx) using SCROLL key and confirm the selection by pressing ENTER key.
3.	Select the type of transmission (TRANSMIT TYPE), according to individual built sheet of the beacon: - With AR6xx no other choice than CMD ONLY. - With RT6xx, if function is to be activated by a single direct word sequence, select CMD ONLY by using SCROLL key. Then confirm the selection by pressing ENTER key.
4.	Enter the hexadecimal code (CODE CMD) using the alphanumerical keys and confirm by pressing ENTER key.
5.	Indicate to the TT801 which type of command code will be sent to the beacon - For RT6xx: Release, Pinger ON, Telemetry (Diagnostic), Other CMD - For AR6xx: Release or Other CMD (for INT/Ranging) Selection is made with SCROLL key and confirmation with ENTER key. The DATABASE RX / SAVE display opens.
6.	Select the type of the information to display (RANGING): DISTANCE or Elapsed TIME using SCROLL key and press ENTER key to confirm. Upon completion of the operation, TT801 displays the message " STORAGE COMPLETED " and return to the MAIN MENU display.
7.	End of procedure.

III.4.5 DELETING AN EXISTING CONFIGURATION

The access to this sub-menu is allowed by pressing **2** key in **TX / RX settings** menu and then by pressing **3** key in the **DATABASE** menu. A window opens with the first instrument available in the database as in following examples:

DATABASE / ERASE		Step 1	DATABASE / ERASE	
▼			▼	
IDENT :	005		IDENT :	012
TYPE :	RT8xx		TYPE :	RT8xx
TRANSMIT TYPE :	ARM + CMD		TRANSMIT TYPE :	IIF
CODE ARM :	DCFE		FREQ :	12 kHz
CMD :	DCFF		RECURRENCE :	AUTO
RECURRENCE :	MANU		RATE :	2 sec.
COMMAND :	RELEASE		SCROLL = CHOICE	
SCROLL = CHOICE			ENTER = VALID	
ENTER = VALID			ENTER = VALID	

Procedure

Step	Action
------	--------

1. Select the new device to be loaded with **SCROLL** key.

TT801 displays the main parameters of the instruments selected.

To correct the ID, use the **CLEAR** key:

A blank window opens with requirement for an ID number.

You must then type in an existing saved ID number and confirm by pressing the **ENTER** key to display the configuration of the selected instrument.

*If the selection is not valid, the message “ID NOT REGISTERED” is prompted and you are required to exit (**HOME** key) or to type in a new ID number.*

Then, press **ENTER** key to confirm the selection

At completion of the operation, TT801 displays the message “ERASE COMPLETED” and return to the MAIN MENU display.

2. End of procedure.



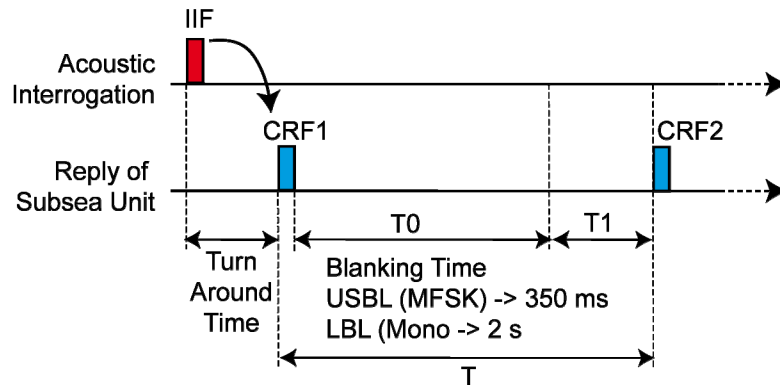
- The instrument with a **null ('0') Identifier (IDENT)** cannot be deleted because it is used as a template with factory default parameters.
- If you press the **ENTER** key after having selected the null ID, the message “**NOT POSSIBLE TO ERASE**” is displayed with no action and the main menu opens.

III.4.6 CASE OF PRESSURE MEASUREMENT

The Keller pressure sensor and the associated telemetry are optional on the transponders. This telemetry pressure measurement is activated upon:

- **Transponder or Responder positioning interrogation** sent by the TT801: The option PRESSURE TELEMETRY ON should be activated in the transponder. The pressure telemetry cycle is added or not after the positioning transmission depending on the status of the Telemetry mode (this mode is activated by the Telemetry ON and terminated by the Telemetry OFF commands from the acoustic remote control). Pressure data can also be forwarded through RS232 serial link if the responder option is available.
- **Remote control** (specific remote control command). This function allows the TT8 to send the PRESSURE command and to receive by acoustic link, the data from an optional pressure sensor on the transponder.

Transponder Mode



Remote Control Mode

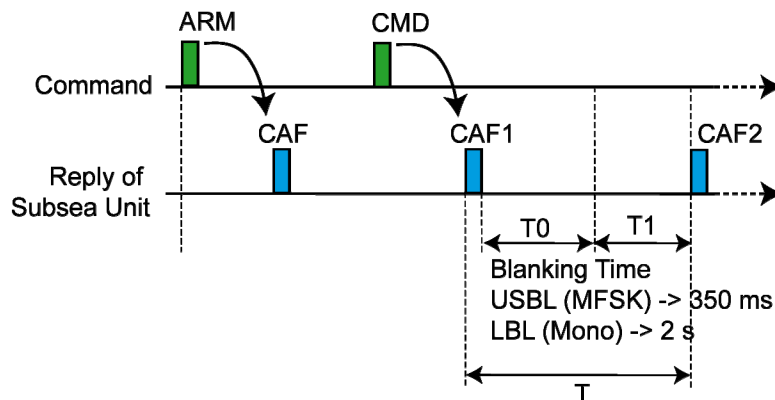


Figure 9 - Pressure measurement diagrams in the two interrogation modes

Pressure Calculation:

Case of a LBL configuration (if CAF or CRF is Tonal or MFSK-BF)

The 12 bits of the measurement are totally transmitted by pulse to pulse time modulation.

T1: time coding of the value of the 12 bits of the measurement

T0: minimum separation between pulses (2 s by default)

K0: time resolution of the coding (2.442 ms by default)

The value of the measurement M is given by:

$$M = (T - T0) / K0$$

The coding unit of pressure measurement is 2 dbar:

$$\text{Pressure} = 2 \times M \text{ (in dbar)}$$

Case of a USBL configuration (if CAF or CRF is MFSK-POSI)

The 12 bits of the measurement are partially transmitted by pulse to pulse time modulation state and partially by MFSK Costas code state.

T1: time coding of the value of the 8 most significant bits of the measurement

T0: minimum separation between acoustic pulses (350 ms by default)

K0: time resolution of the coding so (2.5 ms by default)

$$M1 = (T - T0) / K0$$

CAF2 / CRF2: MFSK pulse for the 4 least significant bits of the measurement

M2: number of permutation CAF2 and CRF2 (0 à 15)

The value of the measurement M is given by:

$$M = 16 \times M1 + M2$$

The coding unit used to transmit the pressure measurement is 2 dbar

$$\text{Pressure} = 2 \times M \text{ (in dbar)}$$

Pressure Protocol

The NMEA pressure string sent via serial link is:

\$PMEVL,XP,MEA,8,CCCC*CK<CR><LF>

XP: product number

CCCC: pressure measurement in dbar

CK: checksum



THE PRESSURE TELEMETRY BY TT801 UNIT IS ONLY POSSIBLE IF THE REPLY OF THE SUBSEA TRANSPONDER IS SET TO TONAL. CHECK IF THE TRANSPONDER FITTED WITH A PRESSURE SENSOR IS CONFIGURED IN TONAL.

III.4.6.1 Procedure in Transponder or Responder Mode

The Figure 10 sums up the procedure to perform in IMMEDIATE TRANSMIT menu. Follow the procedure described in the section III.4.2.1 with the specific parameters:

- Step 2: Choose IIF
- Step 7: Choose MONO, then the PRESSURE MEAS option appears
- Step 9: Set the PRESSURE MEAS to ON

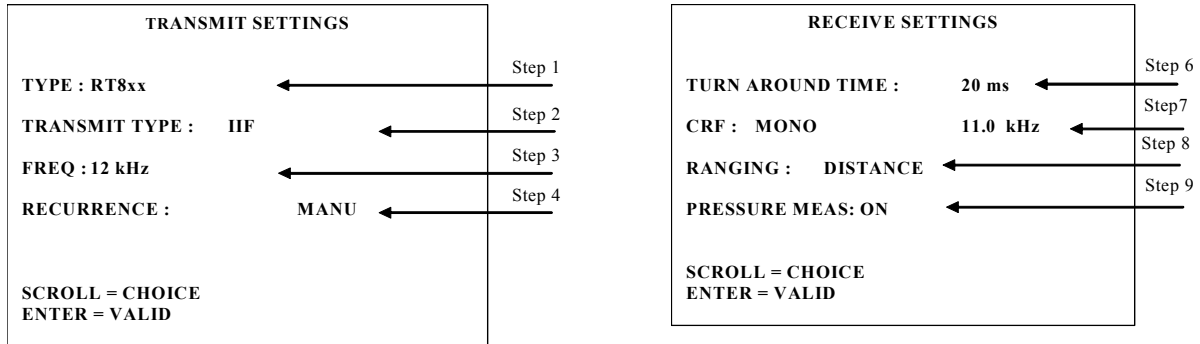


Figure 10 - Displays of Pressure Measurement Configuration

The pressure measurement can also be configured by saving a configuration. Refer to the section III.4.4.1 to find the procedure to follow with the same specific parameters as described above.



If the automatic recurrence is selected for pressure measurement interrogations, it is recommended to evaluate the total duration of the sequence before choosing the period of recurrence.

Result

Upon receipt of an IIF pulse, the beacon transmits two pulses at CRF frequency and the TT801 displays:

```

TYPE : RT8xx
CMD : IIF / PRESSURE MEAS.
FREQ : 12.0 kHz
RECURRENCE : MANU
TURN AROUND TIME : 20 ms
MONO 11.0 kHz      DISTANCE
PRESSURE MEAS : 8 dBars
ENTER = TRANSMIT
    
```

Figure 11 - Display of measured pressure returned

III.4.6.2 Procedure in Remote Control Mode

Procedure

Follow the procedure described in the section III.4.2.2 with the specific parameters:

- Step 2: choose the ARM + CMD option
- Step 4: select the PRESSURE command

Result

Upon receipt of a PRESSURE command, the beacon transmits two pulses at CAF frequency and the TT801 displays the pressure result.

III.4.7 CASE OF TEMPERATURE MEASUREMENT

With transponders equipped with a temperature sensor, it is possible to measure the temperature by telemetry. Temperature measurement is triggered by the TEMPERATURE command. When receiving the command, the beacon transmits a first monochromatic frequency CAF, followed by a second identical CAF.

The graph on Figure 12 illustrates the acoustic transmission of temperature measurement.

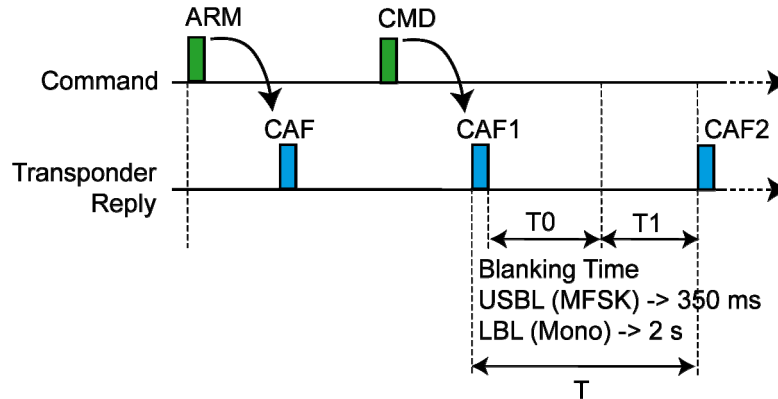


Figure 12 - Temperature Measurement Diagram

Tonal CAF or MFSK-BF

Temperature Calculation in the case of a LBL configuration (Tonal or MFSK-BF CAF)

The 12 bits of the measurement are totally transmitted by pulse to pulse time modulation.

T1: time coding of the value of the 12 bits of the measurement

T0: minimum separation between pulses so 2 s

K0: time resolution of the coding so 2.442 ms

The value of the measurement M1 is given by:

$$M = (T1 - T0) / K0$$

The coding unit temperature measurement is 0.03418 °C

$$\text{Temperature (°C)} = -10 + (M \times 70) / 2048$$

POSI-MFSK CAF

Temperature Calculation in the case of a USBL configuration (POSI-MFSK CAF)

The 12 bits of the measurement are partially transmitted by pulse to pulse time modulation state and partially by MFSK Costas code state.

K0: time resolution of the coding so 2.5 ms

T1: time coding of the value of the 12 most significant bits of the measurement

T0: minimum separation between acoustic pulses: 350 ms

$$M1 = (T1 - T0) / K0$$

CAF2: MFSK pulse for the 4 least significant bits of the measurement

M2: number of permutation CAF2 and CRF2 (0 à 15)

The value of the measurement M is given by:

$$M = 16 \times M1 + M2$$

The coding unit used to transmit the temperature measurement is 0.03418 °C

$$\text{Temperature (°C)} = -10 + (M \times 70) / 2048$$

Procedure Follow the procedure described in the section III.4.2.2 with the specific parameters:

- Step 2: choose the ARM + CMD option
- Step 4: select the TEMPERATURE command

Result Upon receipt of a TEMPERATURE command, the beacon transmits two pulses at CAF frequency and the TT801 displays the temperature result.

III.5 Transmission and Data Reception Display

The TRANSMIT / RECEIVE display is normally used to transmit in water the last command initiated by the operator. The TT801 opens automatically the TRANSMIT/RECEIVE display after a valid selection through:

- IMMEDIATE TRANSMIT sub-menu
- **DATABASE/LOAD** sub-menu

The TT801 opens a TRANSMIT/RECEIVE window such as in the following examples:

After **IMMEDIATE TRANSMIT**:

TYPE : RT8xx	
CMD : PINGER ON	STD
CODE ARM : DCFE	CMD : DCFF
MONO 12.0 KHZ	DISTANCE
ENTER = TRANSMIT	

After **DATABASE/LOAD**

TYPE : RT8xx	IDENT: 100
CMD : RELEASE	STD
CODE ARM : DCFE	CMD : DC55
MONO 12.0 KHZ	DISTANCE
ENTER = TRANSMIT	

The display sums up the type of instrument to be remotely controlled, the command and associated parameters, the TT801 receiver channel setup and the expected data.



The slant range posted up by TT801 in “DISTANCE” mode is derived using a sound speed in water of 1500 m/s.



THE KEYBOARD IS INHIBITED FOR 4 SECONDS DURING IIF, CIF, FR1 OR FR2 INTERROGATION OR ALL ALONG AN **ARM + CMD** COMMAND TRANSMISSION.

PRESSING ANY KEY DURING DATA RECEPTION STOPS THIS MODE.



The acronym ‘STD’ for a FSK transmission means that standard iXBlue timings are used.

The ‘Transmit/Receive’ algorithm for a remote control unit is described in appendix A.

The operator is required to proceed with the transmission as per the following procedure:

Procedure

Step	Action
1.	Transmit the configuration by pressing ENTER key. The same transmission can be sent again by pressing ENTER key. <i>After the acoustic emission of the command, the message “RECEIVING DATA” is displayed on the last line of the window, until detection of the CAF (FT0) acoustic acknowledgement.</i> <i>On the TT801 screen will be displayed the response of the remote beacon with the message “REPLY OK.” and the distance in meter between TT801 and the beacon (or elapsed time in milliseconds) is displayed.</i> <i>Should no acoustic response happened (or detected), the message “NO ANSWER FROM BEACON” is displayed after a time-out (15 to 19 seconds depending on command transmitted).</i> <i>For a Release command, after detection of the acknowledgement of the “command receive” issued by the subsea unit, TT801 displays the message “WAITING FOR RELEASE” until completion of the operation. Upon receipt of the last CAF (acknowledgement for the release execution), “RELEASE OK” message is posted up to confirm success of the operation.</i> <i>For Release + Pinger and Pinger ON commands, the TT801 displays the pinger rate: after receipt of the last CAF acknowledgement (which confirms execution of the command), the settings of the pinger receiver are displayed then the PFR receiver updates the message “RECEIVING DATA” upon detection of the pinger frequency and returns the pinger rate measured with the message “PINGER Rate: value in second”.</i> <i>For Telemetry (Diagnostic) command, the TT801 displays the tilt (Horizontal / Vertical) and the indication of the battery voltage of the beacon: after receipt of the last CAF acknowledgement (which confirms execution of the command), the message “WAITING FOR DIAGNOSTIC” is posted up until the completion of the status telemetry. Then TT801 displays the status message which comprises the Tilt of beacon (VERTICAL or HORIZONTAL) and the battery voltage (Vbat = value in Volt).</i> <i>For PRESSURE and TEMPERATURE commands, the TT801 displays the measurement result after reception of the second CAF allowing the computation of the actual measurement. The message “WAITING FOR TELEMETRY” is displayed between the transmission of the command and the reception of the second CAF.</i>
2.	To stop the reception, press HOME key. <i>The main menu window opens.</i>
3.	End of procedure.

III.6 Activating the Release

This menu allows a quick access to the various pre programmed release commands stored in the TT801 library (Configuration Data Base).

To proceed with this menu, press **2** key in the main menu and in turn the following window opens with the selection of **the first Release command** available in the instrument's library (DATABASE) of the TT801:

RELEASE	
IDENT :	352
TYPE :	RT8xx
CODE ARM :	04C1
CMD :	0455
MONO :	12 kHz
DISTANCE	
COMMAND :	RELEASE
SCROLL = CHOICE	
ENTER = VALID	

Procedure

Step	Action
------	--------

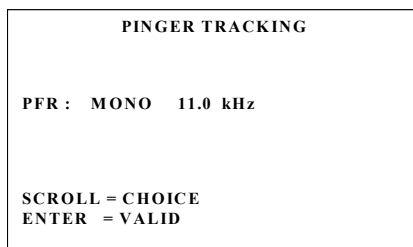
- | | |
|----|---|
| 1. | <p>Select the new device to be loaded with SCROLL key.</p> <p><i>TT801 displays the main parameters of the instruments selected.</i></p> <p>To correct the ID, use the CLEAR key:</p> <p><i>A blank window opens with requirement for an ID number.</i></p> <p>You must then type in an existing saved ID number and confirm by pressing the ENTER key to display the configuration of the selected instrument. If the selection is not valid, the message "ID NOT REGISTRED" or "NO RELEASE COMMAND" is prompted and you are required to exit (HOME key) or to type in a new ID number.</p> <p>Then, press ENTER key to confirm the selection.</p> <p><i>TT801 skips to step 2.</i></p> |
| 2. | <p>The window proposes to accept (transmit) or either to cancel the selection:</p> <ul style="list-style-type: none"> Press ENTER key to transmit the acoustic code(s). Press CLEAR key to cancel the selection. TT801 clears the display and skips to step 1 of this procedure, asking for a new ID number. |
| 3. | End of procedure. |



If you type in an ID number that has been used to save several configurations, the first configuration registered with this ID number will be loaded.

III.7 Activating the Pinger Frequency Detection

This function allows to detect the coherent acoustic pulses coming from a subsea pinger (free running transmitter such as Bottom Finder Pinger) and to listen to the noise generated at the remote dunking transducer (by using the Headphones feature) and. The access to this menu is allowed by pressing **3** key in the main menu.



Procedure

Step	Action
1.	Select the type of PFR (Pinger Frequency) to be detected (MONO or MFSK) then select the frequency (using SCROLL key) or the MFSK code number (type in the code number) and confirm by pressing ENTER key. <i>Then TT801 is waiting for acoustic pulses and the message “RECEIVING DATA” is displayed on the last line of the LCD screen.</i> <i>Each time an acoustic signal matching the TT801 receiver is detected, a dot “.” is displayed and the cursor pushed away to next character position. The received PFR rate is then measured and displayed as “Pinger Rate: xxx”, with xxx in seconds.</i>
2.	You can use the headphones to detect and follow acoustically the pinger frequency.
3.	End of procedure.

III.8 Activating the Options

Reach the menu **OPTIONS** by pressing key **4** in the main menu. The following window opens:

OPTIONS	
LANGUAGE :	ENGLISH
BACKLIGHT :	ON
AUDIO :	ON
BATTERY VOLTAGE:	24.5 V
SCROLL =	CHOICE
ENTER =	VALID

This menu enables the operator to:

- Choose the interface language: only two languages are offered: French and English. English is the default language at power up.
- To turn on/off the display backlighting feature: by default, backlighting is valid. Backlighting mode being activated (ON) helps to read data in very poor visibility conditions.
- To check for battery voltage: Battery voltage is displayed in Volt and refreshed each time the user comes to this menu.
- To activate the command beep: This function enables or disables the “BEEP” audio indicator, each time a character is typed in from the keyboard.



Provisions have been made for a RS232/RS422 I/O port on. But at the time being, this menu does not support RS232/422 link. RS232 link is set by default.

Procedure

Step	Action
1.	To swap from ENGLISH to FRENCH language (or vice versa), use SCROLL key and press ENTER key to validate. <i>After exit the OPTION menu, the various menus are posted up with the chosen language.</i>
2.	To turn ON/OFF the backlighting mode: press SCROLL key to select the active state of the backlighting function (ON or OFF), then press ENTER key to accept.
3.	To activate the command beep: scroll the different suggestions with SCROLL key and validate it by pressing ENTER . <i>The main menu opens, according to the new options.</i>
4.	End of procedure.

III.9 Using the Serial Link

III.9.1 PRINCIPLE

A serial link plug is present on the front panel of the TT801 cabinet. See section I.7 for the description of the RS232/422 plug. This plug allows the operator to get access to data measured by the TT801 unit (Range, Diagnostic Data, Pressure and Temperature measurement, Acknowledgement of Release execution...).

When the TT801 receives a signal from the Acoustic Release/Transponder, it displays the results on its screen and simultaneously transmits strings containing the same results via the serial link. The format of the strings is based on the NMEA 0183 standard.

III.9.2 SERIAL LINK PARAMETERS

Each message is composed of a maximum of 79 ASCII printable characters (\$20 to \$7F) with a front and rear character. This message is sent on the following format:

- 4800 bauds
- 1 start bit
- 8 bits without parity (MSB = 0)
- 1 stop bit

The following parameters have a significant meaning:

- \$ Message start
- , Fields separator
- * Before the checksum
- <CR><LF> Message end

The checksum is computed by using a exclusive OR on the total set of bytes between the \$ and * characters without including these characters. The hexadecimal value of the checksum byte is sent in a two characters format.

III.9.3 MESSAGE STRUCTURE

A message is typically composed of the following structure:

\$PMEVL,XP,RES,AAA,BBB,YP,CMD,xxxxxx *CK<CR><LF>

- **XP** Serial number of the TT801 remote control unit
- **RES** ID of the string (**RES**: Measure REsult ...)
- **AAA** Type of Result data (See Table 7 for more details)
- **BBB** Type of interrogated beacon
- **YP** Serial number of the interrogated acoustic device, **IMM** if the command is IMMEDIATELY sent without saving the configuration
- **CMD** Type of transmitted command (**IIF**, **CMD ONLY**)
- **xxxxxx** Result depending of the type of transmitted data, error cases (no answer from the beacon) are followed by the value '-1'

Depending of the command, the acoustic answer **AAA** can have different meanings.

Table 7 – Correspondence between Commands and Answers

	Chronology	AAA	xxxxxx
CIF / IIF (PRESSURE OFF)	-	RNG	Measure in m
IIF (PRESSURE ON)	CRF ₁	RNG	Measure in m
	CRF ₂	PRE	Measure in dBars
TCMDE RELEASE	CAF ₁	RNG	Measure in m
	CAF ₂	RNG	Measure in m
	CAF ₃	REL	OK
TCMDE DIAGNOSTIC	CAF ₁	RNG	Measure in m
	CAF ₂	RNG	Measure in m
	CAF ₃	DIA	Measure in mV
TCMDE PRESSURE	CAF ₁	RNG	Measure in m
	CAF ₂	RNG	Measure in m
	CAF ₃	DIA	Measure in dBars
TCMDE TEMPERATURE	CAF ₁	RNG	Measure in m
	CAF ₂	RNG	Measure in m
	CAF ₃	DIA	Measure in °C
OTHER TCMDE	CAF ₁	RNG	Measure in m
	CAF ₂	RNG	Measure in m

III.9.3.1 Transponder Command Type

The answers to the commands IIF (FR2) and CIF (FR1) look like:

\$PMEVL,XP,RES,RNG,RT8,YP,CIF,3250 *CK<CR><LF>

3250: Result giving the distance in m computed between the operator and the interrogated acoustic device (this result includes the transponder transmitting delay for a sound speed of 1500 m/s).

In the case where the Pressure Measure is activated, the two acoustic answers of the transponder generate the following strings on the serial link:

Reception CRF₁: **\$PMEVL,XP,RES,RNG,RT8,YP,IIF,3250 *CK<CR><LF>**

Reception CRF₂: **\$PMEVL,XP,RES,PRE,RT8,YP,IIF,3000 *CK<CR><LF>**

3000: Result corresponding to the pressure measure in dbar.

III.9.3.2 Remote Control Unit Command Type

Release The answers to the Release and Release + Pinger commands are:

Reception CAF₁: **\$PMEVL,XP,RES,RNG,RT8,YP, ARM+CMD,3250 *CK<CR><LF>**

Reception CAF₂: **\$PMEVL,XP,RES,RNG,RT8,YP,ARM+CMD,3251 *CK<CR><LF>**

Reception CAF₃: **\$PMEVL,XP,RES,REL,RT8,YP, ARM+CMD,OK *CK<CR><LF>**

Diagnostic The answers to the Diagnostic commands are:

Reception CAF₁: **\$PMEVL,XP,RES,RNG,RT8,YP, ARM+CMD,3250*CK<CR><LF>**

Reception CAF₂: **\$PMEVL,XP,RES,RNG,RT8,YP,ARM+CMD,3251*CK<CR><LF>**

Reception CAF₃: **\$PMEVL,XP,RES,DIA,RT8,YP, ARM+CMD,7630*CK<CR><LF>**

7630: Result corresponding to the Acoustic Release battery voltage in mV.

Other Commands The answers to other types of acoustic commands are:

Reception CAF1: **\$PMEVL,XP,RES,RNG,RT8,YP, ARM+CMD,3250*CK<CR><LF>**

Reception CAF2: **\$PMEVL,XP,RES,RNG,RT8,YP,ARM+CMD,3251*CK<CR><LF>**

3250 or 3251: Range, result corresponding to the distance in m computed from the operator and the acoustic release.

IV MAINTENANCE

IV.1 Object

This section details the preventive and corrective maintenance to perform on TT801.

It also gives clues for troubleshooting.

Servicing an instrument must always be carried out by competent personnel with appropriate training. Generally when dismantling equipment care should always be taken, the disassembly process should be carried out in a clean, free of dust and well-ventilated area.

To...	See...
Know the TT801 Routine Maintenance	IV.2 Routine Maintenance
Know the TT801 Corrective Maintenance	IV.3 Corrective Maintenance
Locate the equipment fault	IV.4 Troubleshooting

IV.2 Routine Maintenance

TT801 has been designed for a simple and easy use. Nevertheless to prolong the life of the unit, regular maintenance must be carried out and logbook kept. In order to overcome the self-discharge during storage period, internal battery charging is also required on regular basis. The following forms describe the maintenance procedure for:

- PM01: Battery Charging
- PM02: Inspection of remote dunking transducer
- PM03: Measurement of Open capacitance of the transducer

System: TT801	Preventive Maintenance	
Sub-system: Battery	Reference: PM01	Page: 1/1
Subject: Charging the Battery	Maintenance level: 1 Periodicity: every 60 hours during operation, every 3 months in normal storage (Storage Temperature > 5°C) Duration: 20 hours (for full charging)	

Equipment AC or DC power cord compatible with TT801 cabinet

Precaution The TT801 must be connected to only TT or TN power supply. Batteries may give off gases when recharged. As a general rule, sealed container should be allowed to vent to atmosphere during the charging process.



WARNING: WHEN USING EXTERNAL DC VOLTAGE FOR BATTERY CHARGING MAKE SURE THAT DC VOLTAGE IS STRICTLY HIGHER THAN 28 VDC.

Preliminary Steps The '**LOW**' LED indicator lights on to report a low battery.
The charging can be performed during TT801 operation. DC voltage must be higher than 28V DC to insure proper battery charge.
No special external charger is required thanks to the TT801 built-in charger.

Procedure

Step	Action
1.	Connect the AC power cord to the mains or DC power cord the DC power supply. <i>The yellow LED indicator AC/DC lights on to indicate battery charge in progress.</i> <i>Full charge of the battery is reached in about 20 hours: the LOW LED indicator for low battery voltage is switched off.</i>
2.	End of procedure.

System: PET801P, PET661S-TT and interconnect cable	Preventive Maintenance	
Sub-system: Dunking Transducer	Reference: PM02	Page: 1/1
Subject: Inspection and Cleaning Transducer Neoprene boot and cable	Maintenance level: 1 Periodicity: after and before each deployment Duration: 10 minutes	

The neoprene boot of the transducer should always be free of grease or oil prior to immersion. The interconnect cable must be fully watertight at least for the section deployed in the water.

Equipment For cleaning use a soapy solution with soft cloth or sponge. Do not use alcohol or petroleum solvents.

Precaution High voltage may exist at both leads of any piezoelectric transducer. It is recommended to short circuit pin **K** and **M** of the SOURIAU 851 08 EC 14-12 P 50 44 mating plug.

Preliminary Steps Transducer should not be connected to the TT801 cabinet.

Procedure

Step	Action
------	--------

1. Acoustic Transducer:

- Immediately after recovery from water, rinses and washes in fresh water.
- Check the neoprene boot is not squashed, split or punctured.
- Check top oil filling screw is in place and absence of oil leakage.
- Get rid of any dust and clean the neoprene boot with a soapy solution to remove trace of grease, oil or contaminants.

If relevant, check the bulkhead connector of the transducer: clean the interior and install a dummy plug before storage.

2. Interconnect cable between transducer and TT801 cabinet:

- Immediately after recovery from water, rinses and washes the cable jacket in fresh water. Take care to get free of water the cable connector(s). Then dry the cable with a soft cloth or sponge.
- Check the cable jacket is not squashed, split.

Check the connector(s) of cable are fully functional.

3. End of procedure.

System: PET801P, PET661S-TT	Preventive Maintenance	
Sub-system: Dunking Transducer	Reference: PM03	Page: 1/1
Subject: Measurement of open capacitance	Maintenance level: 1 Periodicity: annual basis or whenever required Duration: 10 minutes	

A faulty transducer can be detected by measuring its open capacitance. This procedure helps also monitoring the long-term performance of the piezoelectric ceramic included inside the transducer.

Equipment Capacitance meter (ISO-Tech DM9023 or Data Precision model 938 or equivalent) or LCR Bridge.

Precaution High voltage may exist at both leads of any piezoelectric transducer. It is recommended to short circuit pins **K** and **M** of the SOURIAU 851 08 EC 14-12 P 50 44 mating plug to discharge the capacitor of the transducer.

Preliminary Steps Transducer should not be connected to the TT801 circuitry.

Procedure

Step	Action
1.	Discharge the capacitor of the transducer: Short circuit pins K and M of the SOURIAU 851 08 EC 14-12 P 50 44 mating plug mounted on the interconnect cable used between the transducer and TT801 cabinet
2.	Connect the capacitance meter between pins K and M of the SOURIAU 851 08 EC 14-12 P 50 44. A normal transducer (PET801P-30 or PET661S-TT) fitted with 30 m long cable should measure 17.5 ± 2.5 nF . If Transducer is measured without any cable, the capacitance is 14.5 ± 2.0 nF
3.	If any problem contact iXBlue Acoustic Support or After Sale service for replacement or spare part
4.	End of procedure.

IV.3 Corrective Maintenance

It may happen you have to replace the battery or Mains AC fuse and the battery itself as well. The fuses are located on the lateral walls of the power unit (see Figure 13). They are indicated by the icon below that refers to this document for further information.

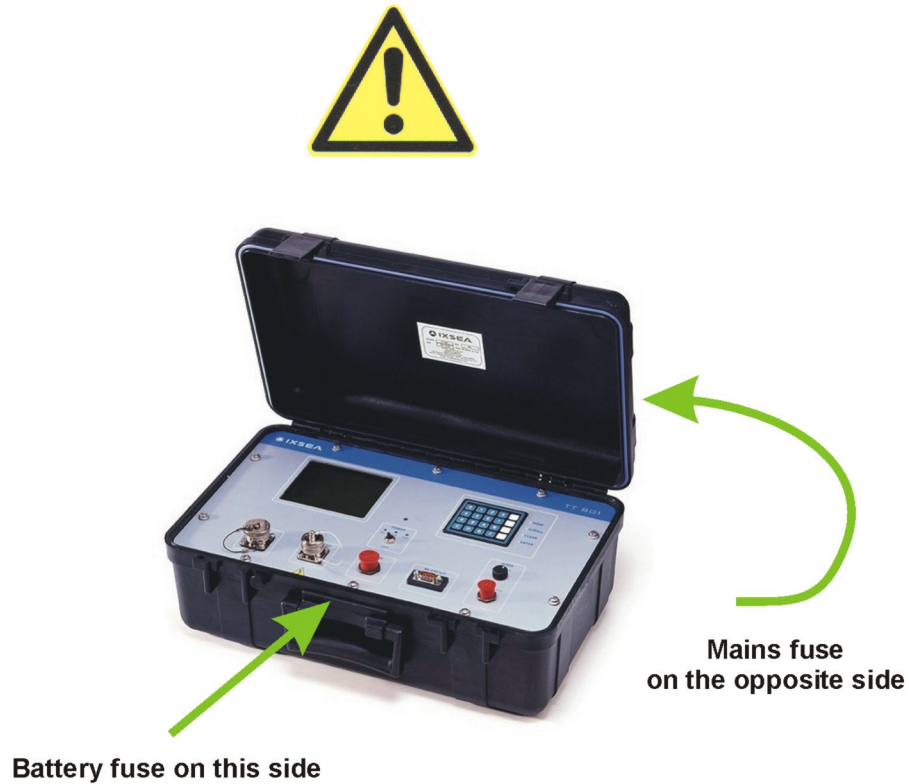


Figure 13 - Fuse locations

Refer to the following procedures to perform the operations

- CM01: Replacing the Battery fuse
- CM02: Replacing the Mains AC fuse
- CM03: Replacement of internal rechargeable Battery

System: TT801	Corrective Maintenance	
Sub-system:	Reference: CM01	Page: 1/2
Subject: Replacing the Battery Fuse	Maintenance level: 2 Duration: less than 10 minutes	

When **ON** LED indicator does not light on and LCD screen remains blank, after switching on the TT801 cabinet. Make sure the battery is charged, before proceeding further.

Equipment
 A flat blade screwdriver (6.5 X 150 mm)
 A battery fuse 5A- 250V (ref.: 5A HPC TDC)

Preliminary Steps
 Switch off the TT801 cabinet. Unplug both cables (acoustic transducer and AC Mains).

Procedure

Step	Action
------	--------

1. Loosen the ten screws that fit the front panel on its case, with the flat blade screwdriver.



2. Remove the front panel from the case:



3. Using the flat blade screwdriver, remove the battery fuse from its stand:



4. Check the old fusible with an ohmmeter and install the new fuse.
5. Fit the front panel back in its case.
6. Refit on the ten screws to fit the front panel on the ABS casing.
7. Switch the TT801 unit on. The ON LED should light on green.
8. End of procedure.

System: TT801	Corrective Maintenance	
Sub-system:	Reference: CM02	Page: 1/2
Subject: Replacing the Mains AC Fuse	Maintenance level: 2 Duration: less than 10 minutes	

When Yellow “AC/DC” LED indicator does not light on when AC mains is connected to TT801 cabinet.

Equipment A flat blade screwdriver (6.5 X 150 mm)
A Mains AC fuse 5 A – 250 V (ref.: 5 A HPC TDC)

Preliminary Steps Switch off the unit.
Unplug both cables (acoustic transducer and AC Mains).

Procedure

Step Action

1. Loosen the ten screws that fit the front panel on its case:



2. Remove the front panel from the case:



3. Using the flat blade screwdriver, remove the Mains AC fuse from its stand:



4. Install the new fuse.
5. Fit the front panel back in its case.
6. Refit the ten screws to fit the front panel on the ABS casing.
7. Connect AC mains power to TT801 cabinet.

The yellow AC/DC LED indicator lights on.

8. End of procedure.

System: TT801	Corrective Maintenance	
Sub-system:	Reference: CM03	Page: 1/3
Subject: Replacing internal rechargeable battery	Maintenance level: 2 Duration: less than 30 minutes	

When When battery does not hold the charge or has got a failure.

Equipment Make sure you have the following items:

- A flat blade screwdriver (6.5 X 150 mm)
- A 7mm tube spanner
- 10 off M4-A4 NYLSTOP Nuts (available from iXBlue After-Sales service)
- 2 off YUASA batteries (12 Volt DC, 3.2 Amp x Hour), p/n: NP 3.2-12FR from YUASA Battery (available from iXBlue After-Sales service)

Precaution If switched ON, TT801 contains hazardous voltages: The transmitter section is capable of generating several thousands volts which can be dangerous if touched. TT801 incorporates capacitors, which are capable of storing large quantities of electrical energy. Avoid inadvertent discharge during dismantling the internal chassis. Once TT801 is switched OFF, storage capacitors are automatically discharged.

Preliminary Steps Before running the procedure:

- Switch off the unit.
- Unplug both cables (acoustic transducer and AC Mains).

Procedure

Step Action

1. Loosen the ten screws that fit the front panel on its case and remove the front panel from the case:



2. Protect the TT801 front panel with a soft cloth and turn it upside down on a table.

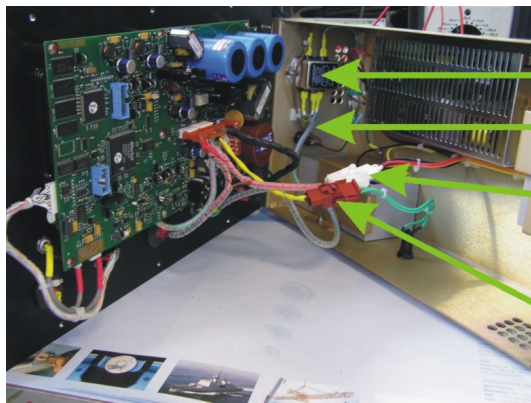
3. Using the 7 mm tube spanner, loosen the 6 off NYLSTOP nuts (3 on each side), which hold the metal chassis onto the TT801 front panel.



Nylstop nuts

System: TT801	Corrective Maintenance	
Sub-system:	Reference: CM03	Page: 2/3
Subject: Replacing internal rechargeable battery	Maintenance level: 2 Duration: less than 30 minutes	

4.
 - Disconnect the 3 cable terminals (blue, brown and yellow/green) from AC Mains filter
 - Disconnect the white flying plug of the Battery pack from the TT801 PCB
 - Disconnect the brown flying of the Battery fuse from the TT801 PCB
 - Separate apart the front panel fitted with the electronic PCB.



AC Mains filter

3xAC-filter cable terminal

White battery flying plug

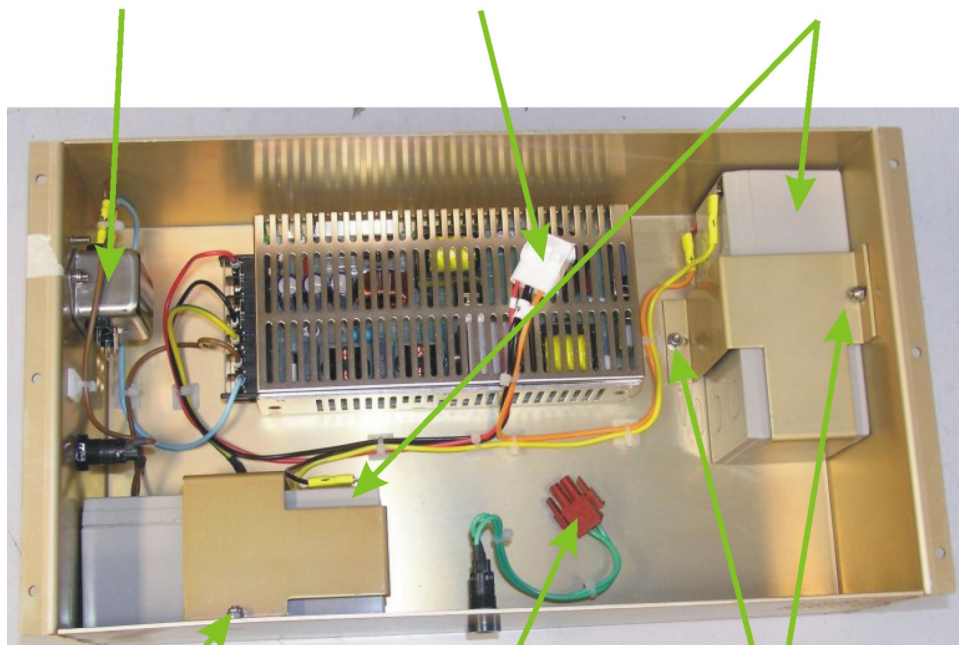
Brown fuse flying plug

5. Withdraw the brackets holding each battery pack onto the metal chassis by loosening the NYLSTOP nuts (2 per battery pack) with the tube spanner.

AC Mains filter

White battery flying plug

2x Battery 12 V 3.2 AH



Battery holder & Nylstop nut

Brown fuse flying plug

Battery holder & Nylstop nuts

System: TT801	Corrective Maintenance	
Sub-system:	Reference: CM03	Page: 3/3
Subject: Replacing internal rechargeable battery	Maintenance level: 2 Duration: less than 30 minutes	

6. Both battery packs are hardwired in series to provide a 24 VDC voltage. Keep track of the wire connection between the two battery packs #1 and #2. Disconnect the orange wire (+24 volt terminal) from battery pack #1, then the black wire (0 volt terminal) from battery pack #2 and then the yellow wire from battery pack #1 and #2.
7.
 - Reconnect the orange, black and yellow wires to the new batteries (check polarity).
 - Insert the new packs and fit each bracket with new M4-A4 NYLSTOP nuts.
 - Tighten the nuts using the 7mm tube spanner.
8. Plug the brown Fuse connector, the white battery connector to the TT801 electronic PCB and then the AC main filter.
9. Re-assemble the metal chassis onto the TT801 front panel with 6 off new M4-A4 NYLSTOP nuts. Tighten the nuts using the 7mm tube spanner.
10. Fit the front panel back in its case.
11. Refit the ten screws to fit the front panel on the ABS casing.
12. Connect AC mains power to TT801 cabinet and charge the new internal battery pack as per the procedure PM01.
13. End of procedure.

IV.4 Troubleshooting

Look at the following table that gives the most common problems.

If you do not find a solution, please contact the iXBlue support (see Warranty and Customer Support).

Problem	Possible causes	Solution(s)
There is no display on the TT801 screen	Power supply problem and/or LCD display failure	1. Make sure the internal battery is charged. 2. Check and change the battery fuse. 3. If no improvement, contact the iXBlue support.
AC/DC yellow indicator does not light on	AC fuse and/or AC charger problem	1. Check that the power supply cable of TT801 unit is correctly plugged. 2. Check that your power supply is delivering tension. 3. Check and change the AC fuse. 4. If no improvement, contact the iXBlue support.
TT801 does not transmit acoustic bursts	Connection problem or Electronic/Transducer failure	1. Check that the transducer cable is correctly connected to the TT801 unit. 2. Check the capacitance of the transducer + cable assembly. 3. If no improvement, contact the iXBlue support.
TT801 does not receive acoustic pulses	Connection problem, Electronic/Transducer failure, Configuration problem or problem with environmental conditions	1. Check that the transducer cable is correctly connected. 2. Check the receiving configuration of the TT801 cabinet: Is the receiving channel set to the expected signal (type of pulse and/or frequency). 3. Is the noise level at the surface dunking transducer too high? Lower the dunking transducer and/or find a new location to deploy the dunking transducer. Make noise and check the headphones. 4. Surface current may be streaming the dunking transducer, such that it is flying at the air-water interface with bubbles. Adding ballast to the dunking transducer may reduce this. 5. A surface thermocline maybe refracting the acknowledgement pulses away from the surface. Lower the dunking transducer below the thermocline layer. 6. Other acoustic system (Echo-sounder,...) may be operating in the same frequency band. 7. Check the capacitance of the transducer + cable assembly 8. Connect the headphones. Knock several times on the transducer and listen to the “noise” received by the transducer. Do you hear something? 9. No, contact the iXBlue support.

iXBlue CONTACT - SUPPORT 24/7 CUSTOMER SUPPORT HELPLINE

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Appendices

A. TECHNICAL SPECIFICATIONS

A.1 Power Supply

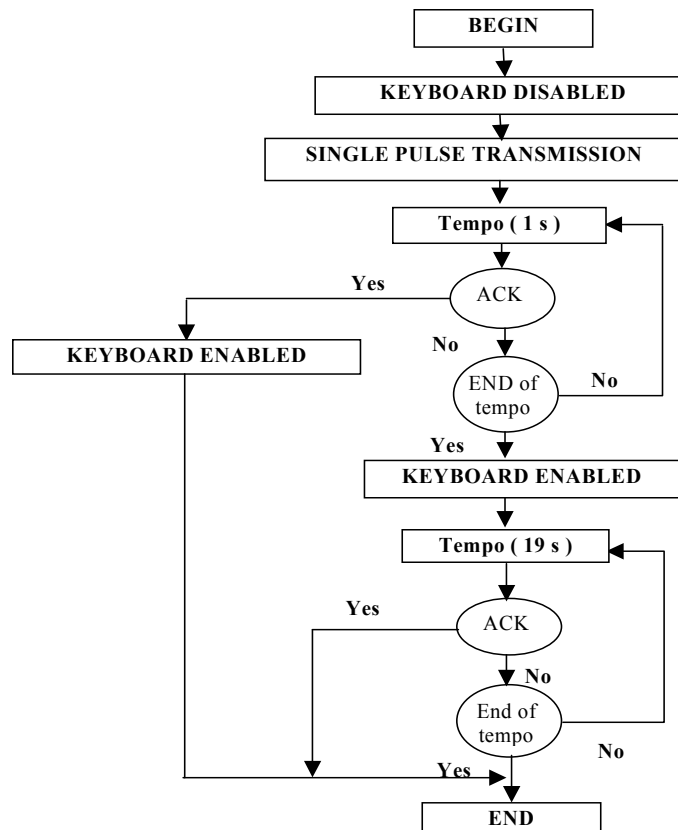
- Internal rechargeable Lead-Acid battery: operational for about 60 hours.
- Standard AC Mains power supply: 115-230V / 50-60 Hz.
- DC power supply: 18-36V. (For battery charging, voltage must be strictly higher than 28VDC)

A.2 Single Pulse Transmission

- Programmable frequencies from 8.5 to 16 kHz (resolution 500 Hz),
- Pulse width: 11 ms,
- Acoustic source level: 190 ± 4 dB ref. $1 \mu\text{Pa}$ at 1m,
- Transducer beam pattern: omni directional ± 3 dB.

Transmit/Receive Algorithm for Single Pulse Transmission:

This one is used for IIF and CIF interrogations.



A.3 Command Codes

- Coding: 2 state, FSK, 8 bit words,
- Number of pulses for each command word: 8,
- Pulse rate: 100 ms,
- Total duration of one command: 711 ms,
- Least significant bit is transmitted first,
- Transmission of ARM and EXECUTION codes through automatic procedure.

Transmit/Receive algorithm for command codes:

This one is used for remote control interrogations.



A.4 Front Panel Features

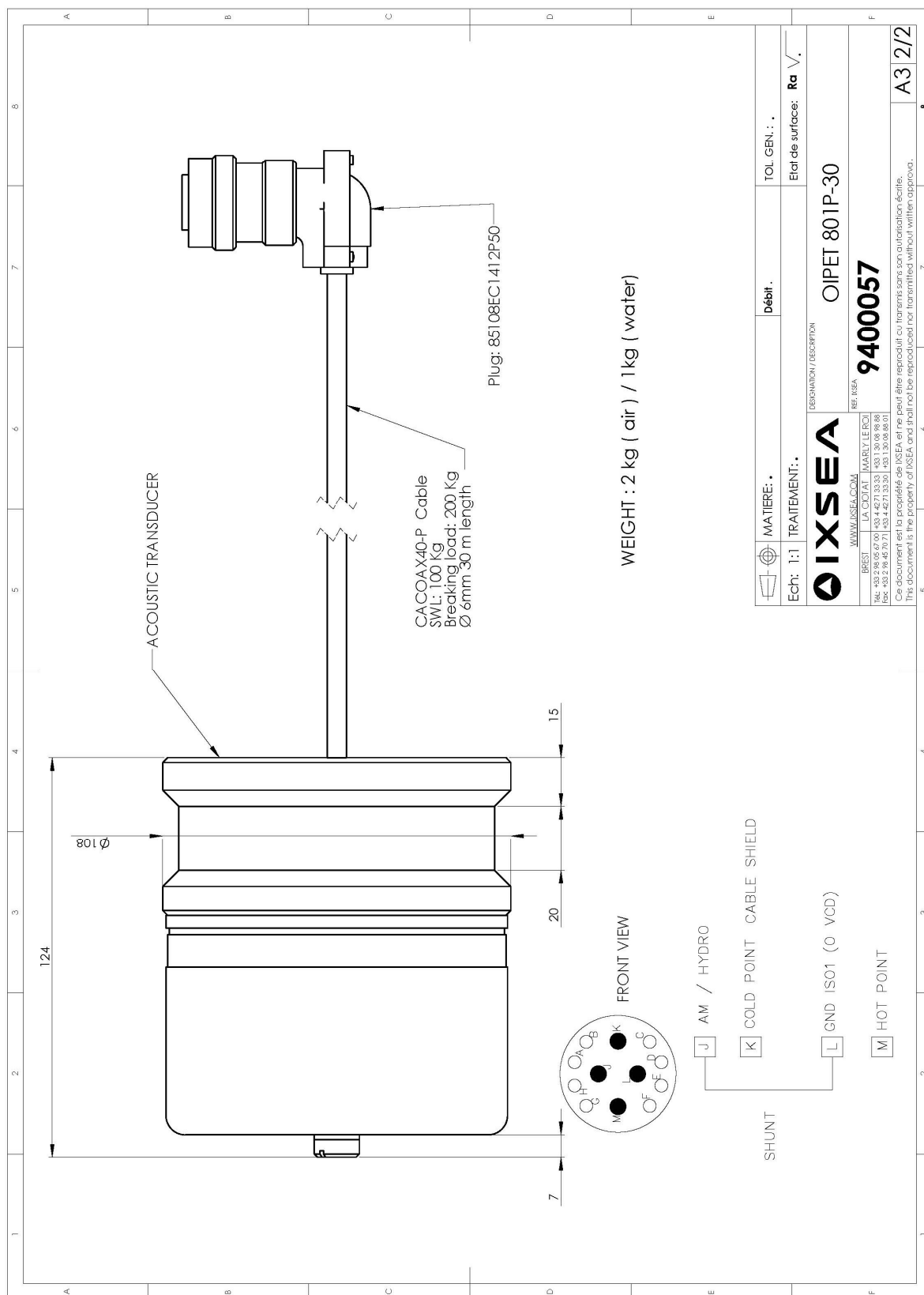
- ON / OFF switch
- SOURIAU 12 way female socket for transducer output (XDC)
- JAEGER 3 way male socket for external 115-230V AC power supply
- SOURIAU 6 way male socket for external 24 VDC power supply
- SOURIAU 4 way female socket for Headphones input
- 9 way “D” male socket for RS232/422 serial link (Provisional)
- **AC/DC** LED indicator for external power supply
- **ON** LED indicator for device's power-up
- **LOW** LED indicator for low battery voltage
- 20-key keyboard for data and command input
- Liquid Crystal Display 16 lines × 20 characters
- Photo resistor for backlighting command
- Audio buzzer

B. FREQUENCY TO DIGIT CONVERSION TABLE

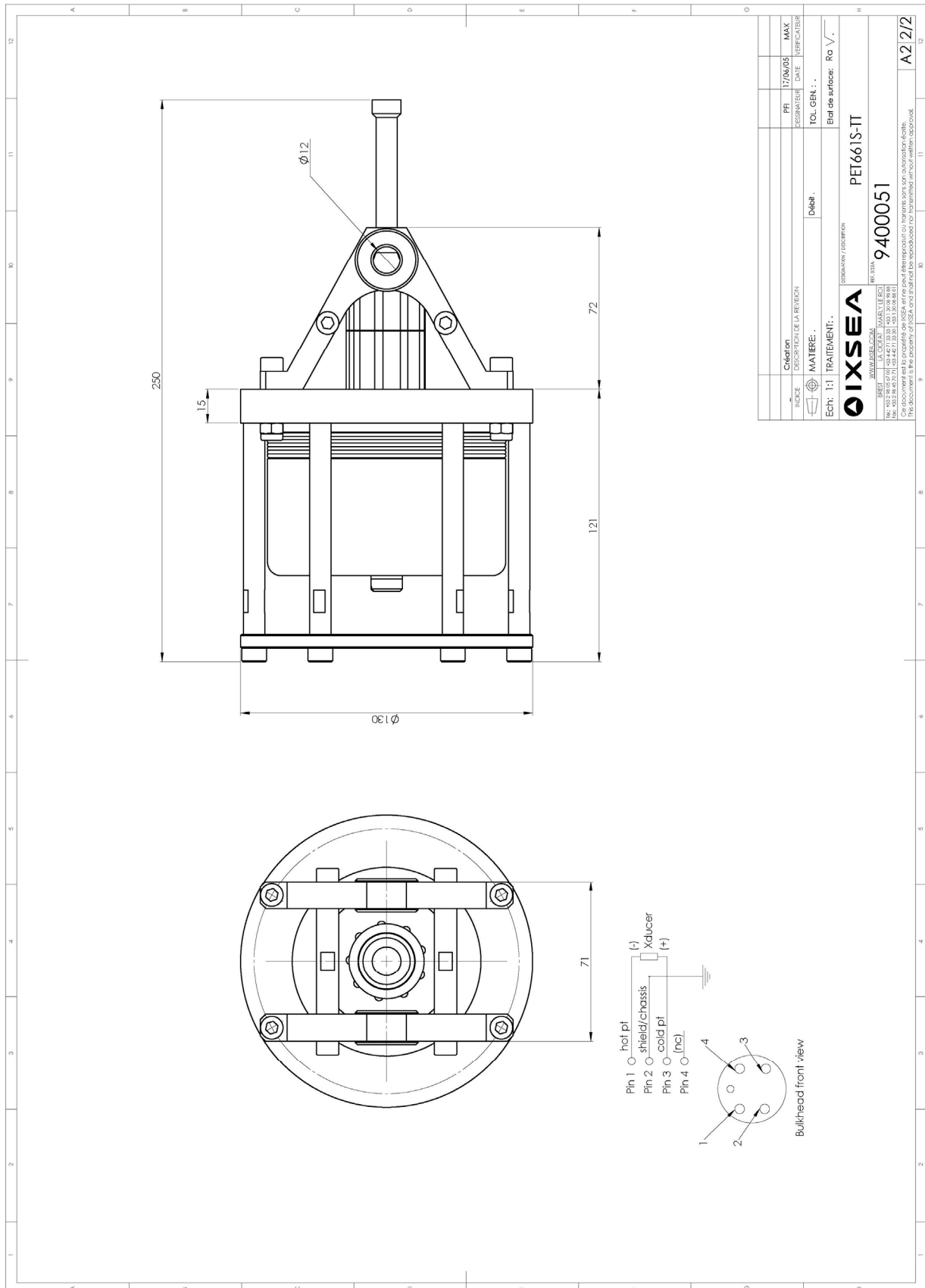
Transmission Frequency	Frequency Code	Bit_0 or Bit_1 Channel
(kHz)	TT301 / TT801	TT201
8.5	0	1
9.0	1	2
9.5	2	3
10.0	3	4
10.5	4	5
11.0	5	6
11.5	6	7
12.0	7	8
12.5	8	9
13.0	9	10
13.5	A	11
14.0	B	12
14.5	C	13
15.0	D	14
15.5	E	15
16.0	F	16

C. DUNKING TRANSDUCERS

C.1 Standard PET801P-30



C.2 Optional PET661S-TT with interconnect cable



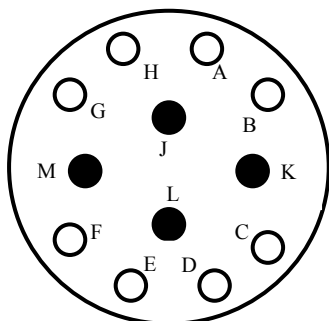
INTERCONNECT CABLE

Between TT701 or TT801 and PET 661 S-TT (iXSea ref : 9300117)

Cable : **AXON ZN2019-MSTM7**
- 7 twisted conducting wires
- Operating voltage : 600 V rms (1000 V rms allowed for a short time)

Length : **30 m**

Connectors :
12 pins (Male) plug SOURIAU series
Ref. :
451 01 EC14-12P (iXSea ref.: P O8EC14-12P)

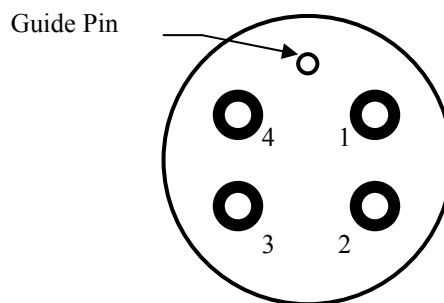


J – AM / Transducer
K – Cold point (yellow) + shield (black)
L – Ground ISO1 (0 V DC)
M – Hot point (brown)

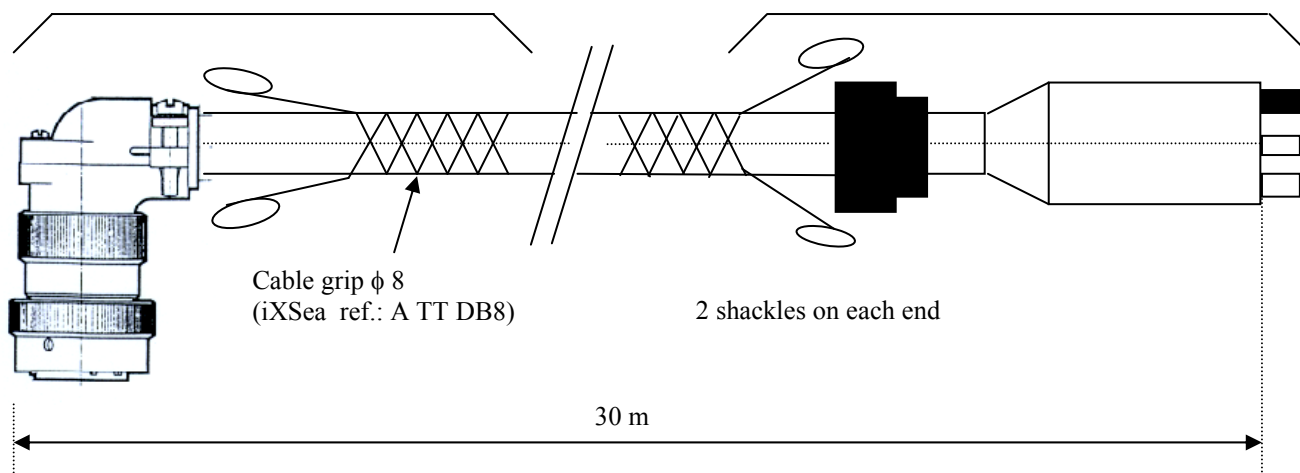
short-circuit between J and L

4 pins (Male) plug SUBCONN series
Ref. :
Plug : **OM4-M (iXSea ref.: P OM4-M)**
Boot : **OMBA (iXSea ref.: P OMBA)**
Locking : **DLSA-M (iXSea ref.: P DLSA-M)**

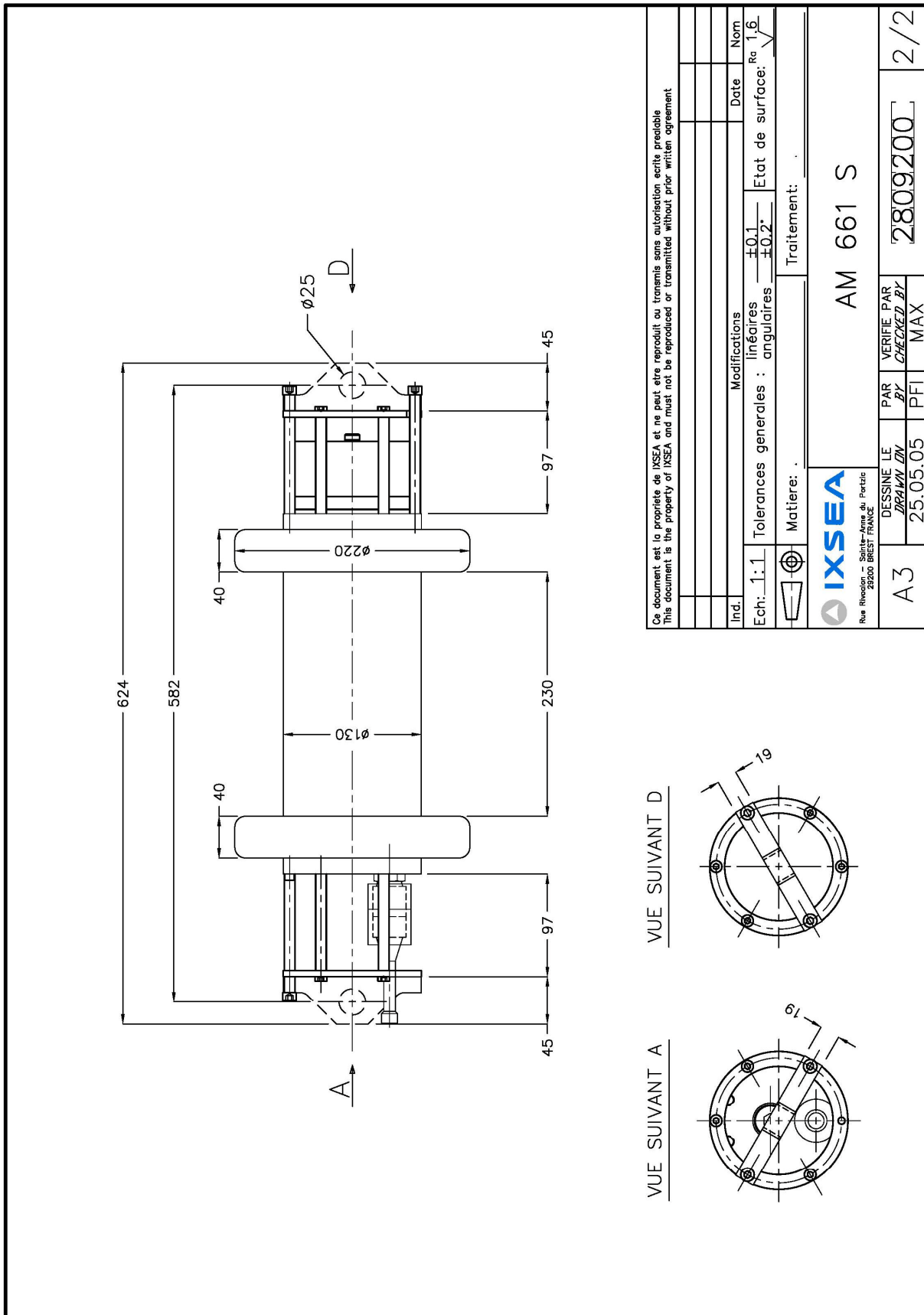
FRONT VIEW



1 - Hot point (brown)
2 - Shield / Chassis
3 - Cold point (yellow)
4 - Not connected (green)



C.3 Optional AM661S Acoustic Module with interconnect cable



INTERCONNECT CABLE

Between TT301B, TT801 and AM661S

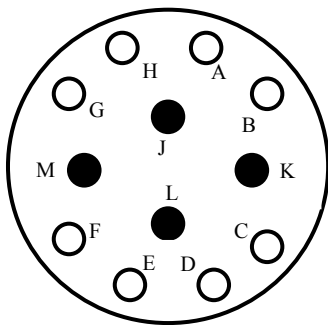
(Ixsea ref: 9300102)

Cable model : **AXON ZN2019-MSTMR7 (Ixsea ref. : CA7-9EP)**
 - 7 twisted conducting wires
 - Operating voltage : 600 V rms (1000 V rms allowed for a short time)

Length : **100 m**

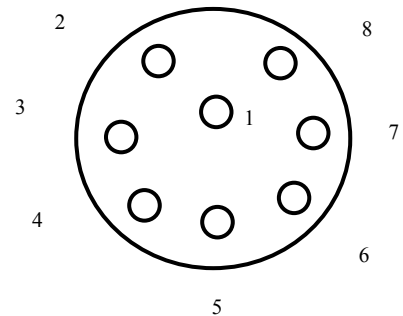
Connections :

12 pins (Male) plug SOURIAU series
851 08EC1412P50 (iXSea ref.: P 08EC14-12P)

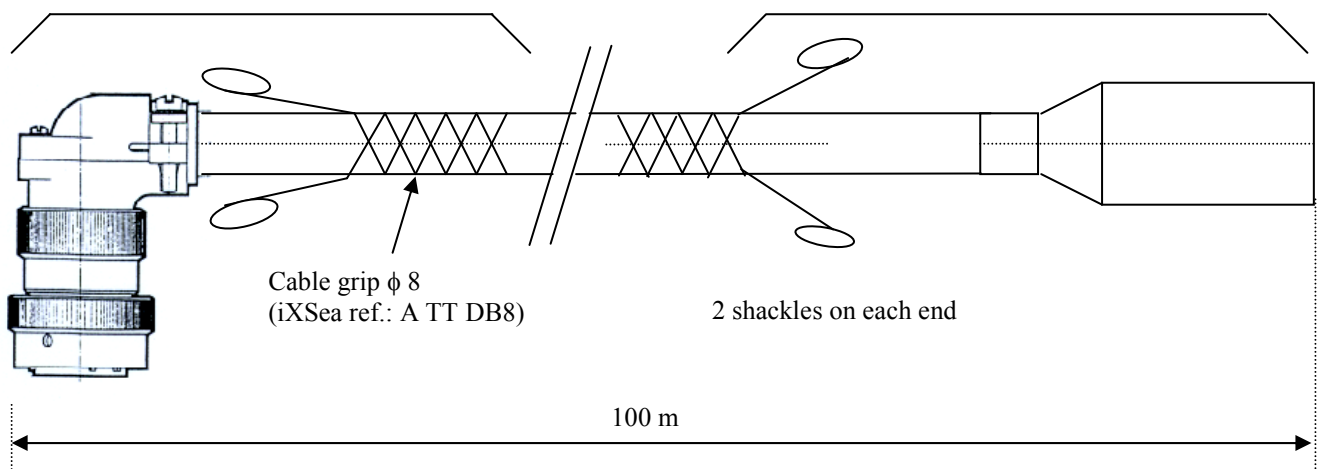


FRONT VIEW

FRONT VIEW
8 pins (female) plug SUBCONN series
 Plug : **OM8F (iXSea ref.: P OM8F)**
 Boot : **OMBB (iXSea ref.: P OMBA)**
 Locking : **DLSB-F (iXSea ref.: P DLSB-F)**



	TT side		AM side
Ground ISO2 Power AM (0V DC)	A	Brown / Marron	1
VAM (+) POWER AM (+48V DC)	B	Red / Rouge	2
LF-PULSE-TR	C	Orange / Orange	3
LF-AM-REC	D	Yellow / Jaune	4
Ground ISO2 Power AM (0V DC)	E	Green / Vert	5
VAM (+) POWER AM (+48V DC)	F	Grey / Gris	7
LF-AM-RET (Cable common Shield)	G	Black / Noir	8
LF-PULSE-RET (Return)	H	Blue / Bleu	6
Not connected	J-K-L-M		



D. EXAMPLES OF BUILD SHEETS

D.1 Example of AR661 Build Sheet

TYPE	:	AR 661 B2S-DDL	Manufacture	:	
S/N	:	698	Customer	:	
P/N	:	301 9200	Agent	:	
Function	:	Acoustic Release	Customer Visa	:	
Modification	:		Affair	:	

TECHNICAL SPECIFICATIONS

ELECTRONIC BOARD				ELECTRONICAL SPECIFICATIONS	
Reference	Rev	Function	S/N	Transmit width	: 10 ms
194 2003	4	AR 6x1 Board Soft V1.0	688	Transmit level	: 190 dB +/-3 dB ref 1µP at 1m
				Transmit frequency	: FT0 = 12.0 kHz
				BIT1 = 12	⇒ FR1 = 14.0 kHz
				BIT0 = 05	⇒ FR2 = 10.5 kHz

FUNCTIONAL SPECIFICATIONS

Function / Code	TT801/TT301	TT201	Sequence	
INT / RANGE	4B13	BIT0 + BIT1 + 13⇒	FT0	
RELEASE	4B14	BIT0 + BIT1 + 14⇒	FT0	⇒ FT0

D.2 Example of RT661 Build Sheet

TYPE	:	RT 661 B2S-DDL	Manufacture	:
S/N	:	486	Customer	:
P/N	:	308 9301	Agent	:
Function	:	Recoverable Transponder	Customer Visa	:
Modification	:		Affair	:

TECHNICAL SPECIFICATIONS

ELECTRONIC BOARD				ELECTRONICAL SPECIFICATIONS	
Reference	Rev	Function	S/N		
194 2002	6	RT 6x1 Board Soft V2.5	552	Transmit width	: 10 ms
				Transmit level	: 191 dB +/-4dB ref 1μPa at 1m
				Transmit delay (↔Turn-around time)	: 15 ms
				Transmit inhibition	: 7.6s
				FT0 = 12.0 kHz	
				FR1 = 09.0 kHz ⇒	FT1 = 14.5 kHz
				FR2 = 14.5 kHz ⇒	FT2 = 09.0 kHz
				BIT1 = 09 ⇒	FR3 = 12.5 kHz
				BIT0 = 03 ⇒	FR4 = 09.5 kHz
				FT4= FT5= FT7 = 12.0 kHz	

FUNCTIONAL SPECIFICATIONS

Function / Code	TT801/TT301	TT201	Sequence		
ON FR1-FR2	2842	BIT0 + BIT1 + 42	FT0		
OFF FR1-FR2-PINGER(ST)	2843	BIT0 + BIT1 + 43	FT0		
RELEASE 1	2844	BIT0 + BIT1 + 44	FT0 ⇒ FT5		
DIAGNOSTIC	2845	BIT0 + BIT1 + 45	FT0 ⇒ FT7	Measure delay : 3s	Verticality Offset : 10s
WINDOW (↔ ARM)	2841	BIT0 + BIT1 + 41	FT0	Wait time : 15s	Active time : 15s
ON FR1-OFF FR2(W)	2883	BIT0 + BIT1 + 83	FT0		
ON FR2-OFF FR1(W)	2884	BIT0 + BIT1 + 84	FT0		
PINGER (W)	2894	BIT0 + BIT1 + 94	FT0 ⇒ FT4	Pulse width : 9.14ms	Recurrence : 1s
(W): Code active only during WINDOW active time					
(ST): Default operating mode at power up					

D.3 Example of AR861 Build Sheet

TYPE	: AR 861 B2S	Date of Manufacture	:
S/N	: 276	Customer	:
P/N	: 392 9100	Representative	:
Function	: Acoustic Release	Job file	:
Modification	:	Customer approval	:

TECHNICAL SPECIFICATIONS

ELECTRONIC BOARD				ELECTRONIC SPECIFICATIONS
<u>Reference</u>	<u>Rev</u>	<u>Function</u>	<u>S/N</u>	
392 2001	3.0	AR 8x1 Board Firmware: PROM (U6) - ET8_V2.2 FPGA (U38) - REC_V1.0/3.3V PROM (U32) - EM_V1.0 FPGA (U33) - EM_V1.0/3.3V	276	Transmit width : 10 ms Transmit level : 191 ± 4 dB ref 1µPa at 1 m Pinger rate : 5 s Pinger duration after release : 3 mn FR0 = 09.0 kHz FR1 = 10.5 kHz CAF = 12.0 kHz PINGER (↔PFR) = 12.0 kHz

FUNCTIONAL SPECIFICATIONS

Function / Code	TT801/ TT701/ TT301	TT201	Sequence
ARM / RANGING	141C	N.A.	⇒ CAF Lock-Out time = 4s Active time = 20s
<u>The following acoustic codes must be preceded by an ARM code</u>			
RELEASE	1455	N.A.	⇒ CAF ⇒ CAF
RELEASE WITH PINGER	1456	N.A.	⇒ CAF ⇒ CAF⇒ PINGER
PINGER ON	1447	N.A.	⇒ CAF ⇒ PINGER
PINGER OFF	1448	N.A.	⇒ CAF
DIAGNOSTIC	1449	N.A.	⇒ CAF ₁ ⇒ CAF ₂
N.A.: Not applicable			

D.4 Example of RT861 Build Sheet

TYPE	:	RT 861 CT-TR	Manufacturing date	:	
S/N	:	305	Customer	:	
P/N	:	386 9501	Representative	:	
Function	:	Recoverable Transponder	Job file	:	
Modification	:		Customer Approval	:	

TECHNICAL SPECIFICATIONS

ELECTRONIC BOARD				ELECTRONIC SPECIFICATIONS
Reference	Rev	Function	S/N	
385 2010	3.0	RT 8x1 Board Soft: PROM (U6) – ET8_V2.2 FPGA (U38) - REC_V1.0/3.3V PROM (U32) - EM_V1.0 FPGA (U33) - EM_V1.0/3.3V	105	Transmit width : 10 ms Transmit level : 191 dB +/-4 dB ref 1μPa at 1 m Transmit delay (↔ Turn-around time) : 20 ms Turnaround delay : 4 s Pinger rate : 5 s Pinger duration after release : 3 mn CIF = 09.0 kHz ⇒ IRF = 13.5 kHz IIF = 13.5 kHz ⇒ CRF = Code 22 FR0 = 08.5 kHz FR1 = 10.0 kHz CAF = 12.0 kHz PINGER (↔ PFR) = 12.0 kHz

FUNCTIONAL SPECIFICATIONS

Function / Code	TT801/TT301	TT201	Sequence	
ARM / RANGING	03B9	N.A.	CAF	Lock-out time = 4s Active time = 20s
<u>The following acoustic codes must be preceded by an ARM code</u>				
RELEASE	ARM + 0355	N.A.	⇒ CAF	⇒ CAF
RELEASE WITH PINGER	ARM + 0356	N.A.	⇒ CAF	⇒ CAF ⇒ PINGER
TRANSP.IIF TRANSP.CIF	ARM + 0341	N.A.	⇒ CAF	
INHIBITION IIF CIF (ST)	ARM + 0342	N.A.	⇒ CAF	
TRANSP.IIF INHIB.CIF	ARM + 0343	N.A.	⇒ CAF	
TRANSP.CIF INHIB.IIF	ARM + 0344	N.A.	⇒ CAF	

PINGER ON	ARM + 0347	N.A.	⇒ CAF ⇒ PINGER
PINGER OFF	ARM + 0348	N.A.	⇒ CAF
DIAGNOSTIC	ARM + 0349	N.A.	⇒ CAF1 ⇒ CAF2
ACOUSTIC RESET	ARM + 034A	N.A.	⇒ CAF
(ST) : Default operating mode at power up			

D.5 Example of AR701 Build Sheet

TYPE	:	AR 701 AE	Date of Manufacture	:	
S/N	:	149	Customer	:	
P/N	:	359 9000	Representative	:	
Function	:	Acoustic Release	Job file	:	

TECHNICAL SPECIFICATIONS

ELECTRONIC BOARD

<u>Reference</u>	<u>Rev</u>	<u>Function</u>	<u>S/N</u>
359 2002	1.1	AR 701 board Firmware : V1.12	149

ELECTRONIC SPECIFICATIONS

FR2	=	14.0 kHz
FR1	=	15.0 kHz

FUNCTIONAL SPECIFICATIONS

Function / Code	TT301/TT701/TT801	TT201
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ARM	BDAD	Not Applicable
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RELEASE	BD06	Not Applicable
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RELEASE code must be preceded by an ARM code.

The lock-out time between ARM code and RELEASE code is **4 s**.

The active time for receiving the RELEASE code is then **20 s**.

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